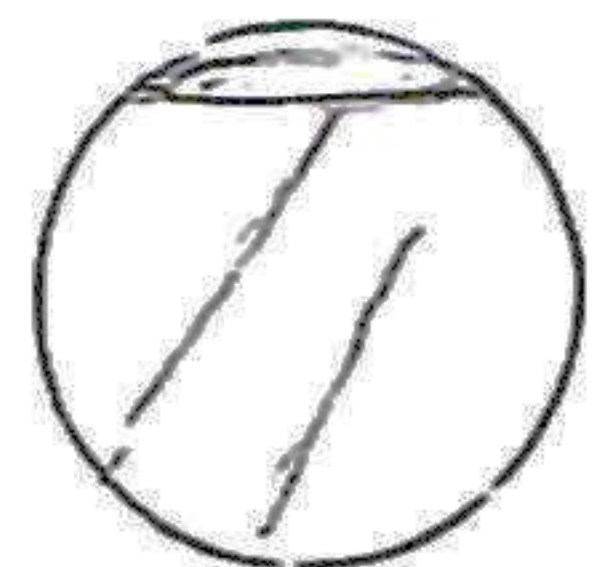


CONCEPT SHEET

UNDERWATER CAVE HOUSE

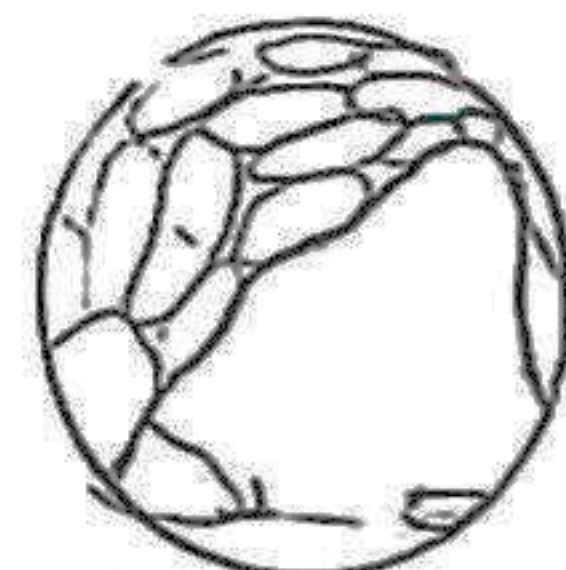
OBSERVATION



SUNLIGHT
PENETRATING
THROUGH WATER



MARINE
LIFE MOVEMENT
AND PATTERNS

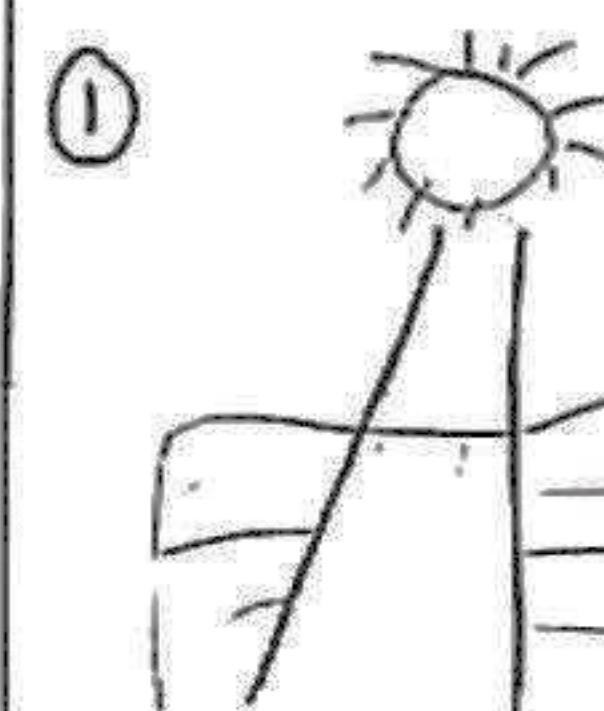


UNDERWATER
CAVES AND
NATURAL VOIDS



CORAL REEF
STRUCTURE AND
ORGANIC FORMS

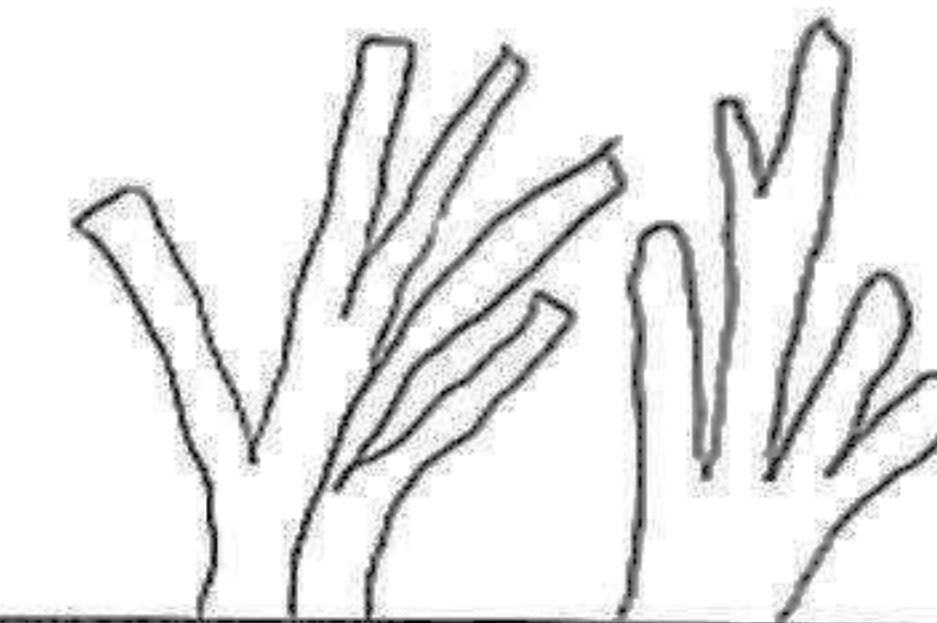
CONCEPT EVOLUTION



1 LIGHT SOURCE



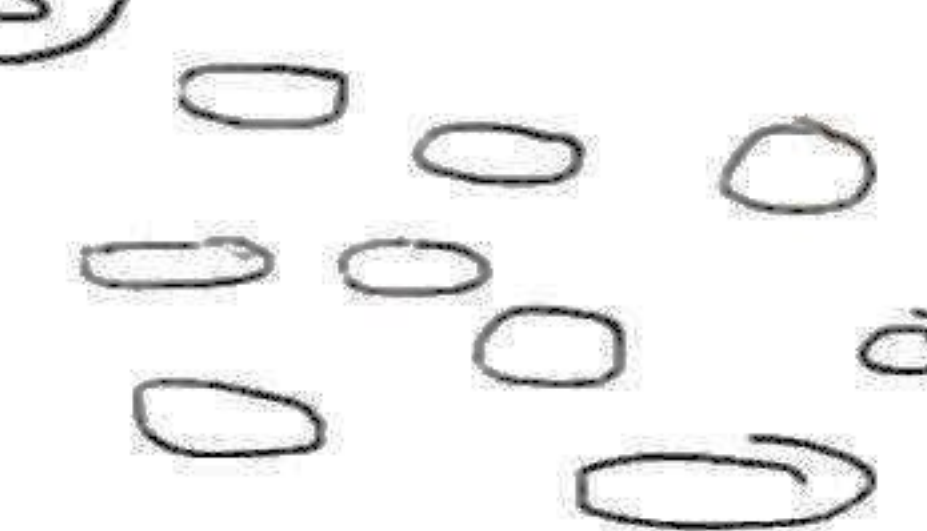
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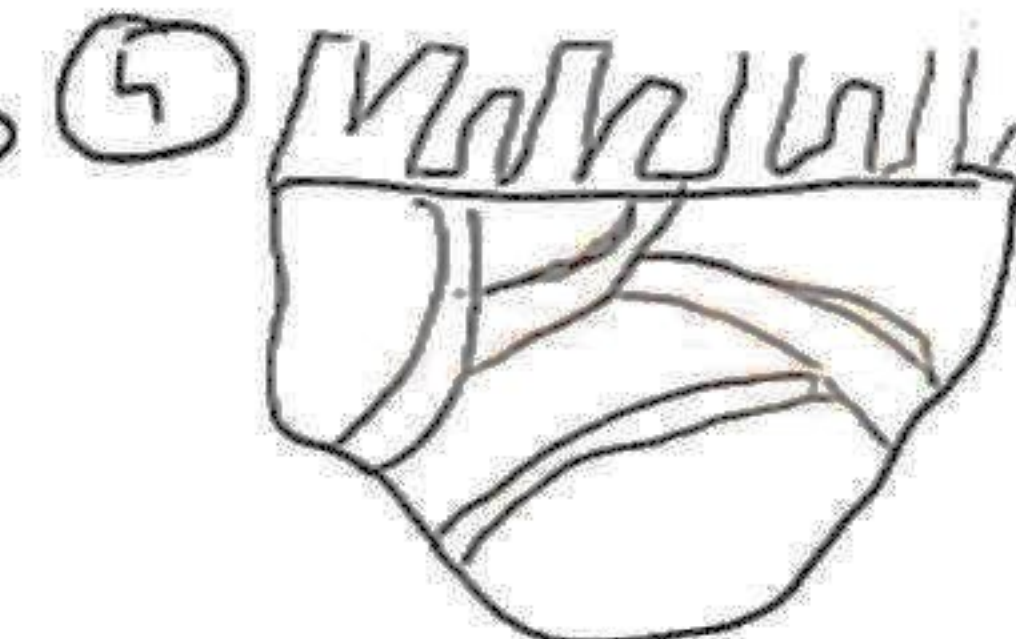
CORAL GROWTH



3



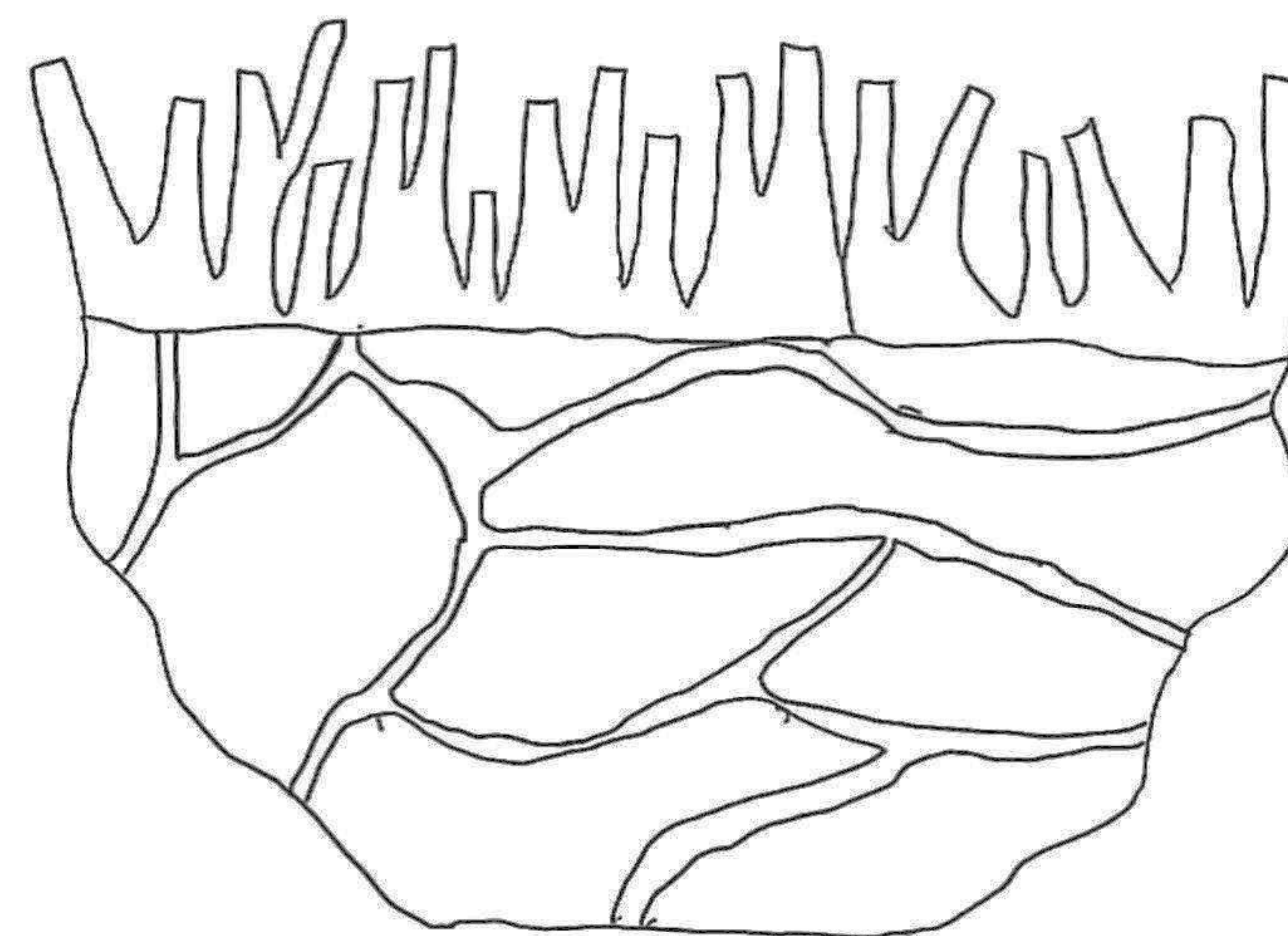
ORGANIC PATTERNS



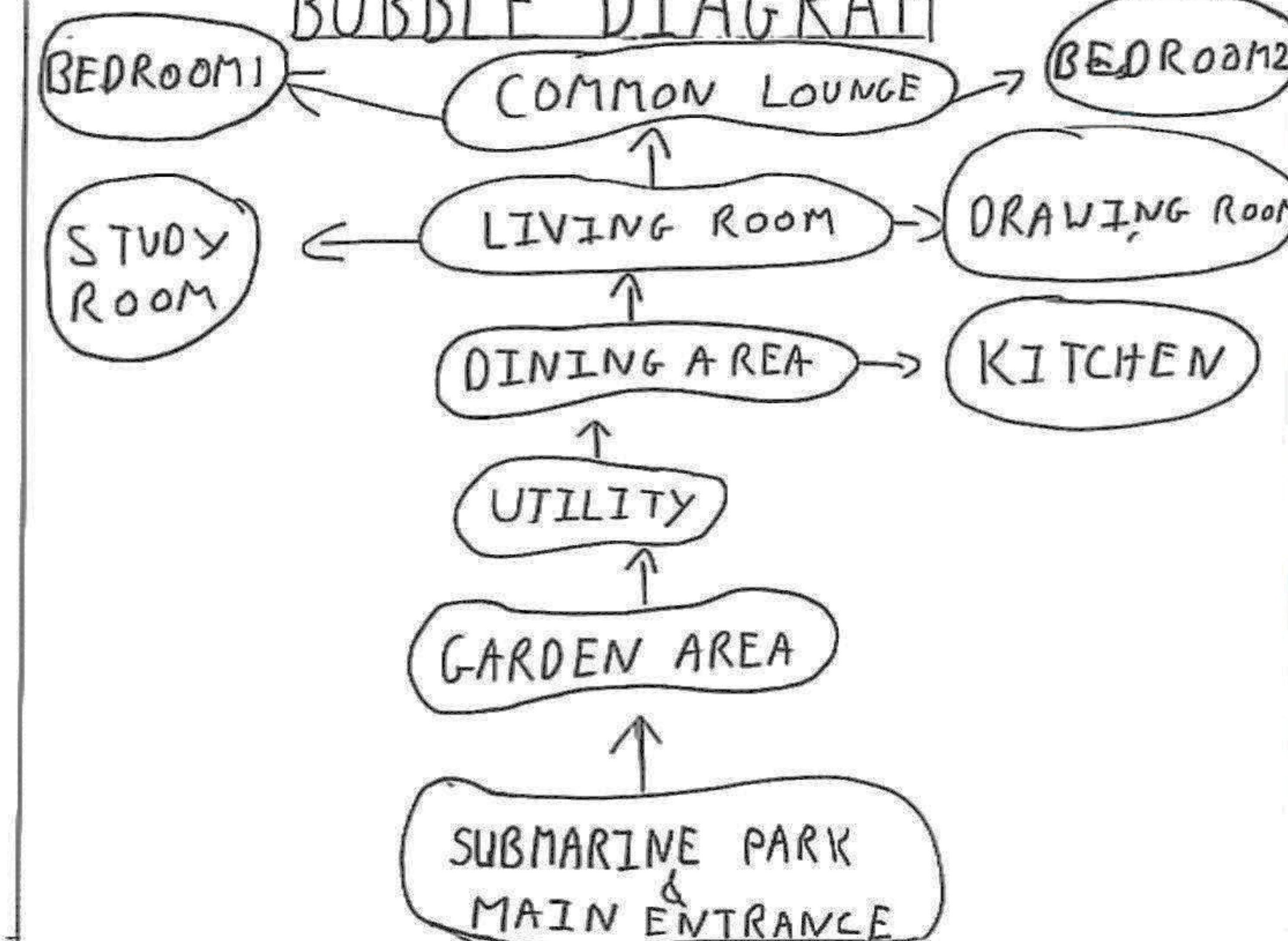
FINAL FORM

MAIN CONCEPT

THE UNDERWATER HOUSE TRANSFORMS
MARINE FORMS INTO ARCHITECTURE BY
CREATING INTERCONNECTED FAMILY
SPACES AROUND A CENTRAL LIGHT VOID.
SUNLIGHT FILTERS THROUGH THE STRUCTURE
SIMILAR TO LIGHT PASSING THROUGH
WATER, WHILE CORAL-INSPIRED FORMS
CREATE A BALANCE BETWEEN PRIVACY
AND INTERACTION.



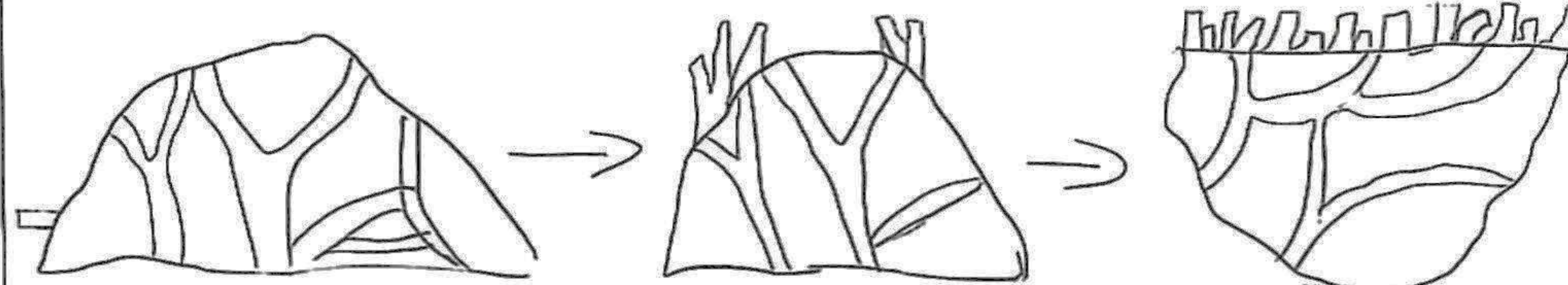
BUBBLE DIAGRAM



MATERIAL PALETTE

- REINFORCED CONCRETE
- TINTED GLASS
- NATURAL STONE FINISH
- CORAL REEFS

FORM DEVELOPMENT

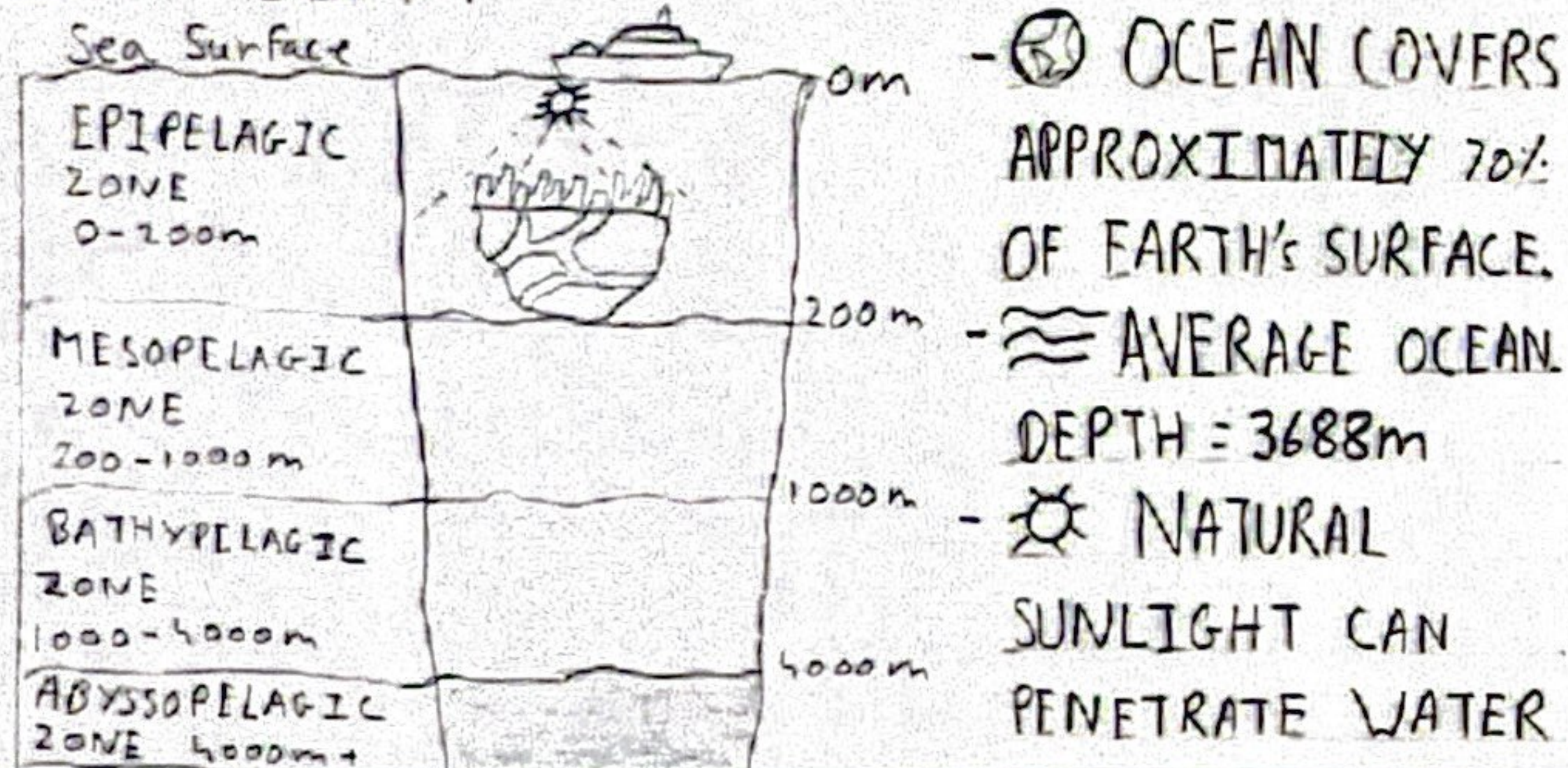


KEYWORDS

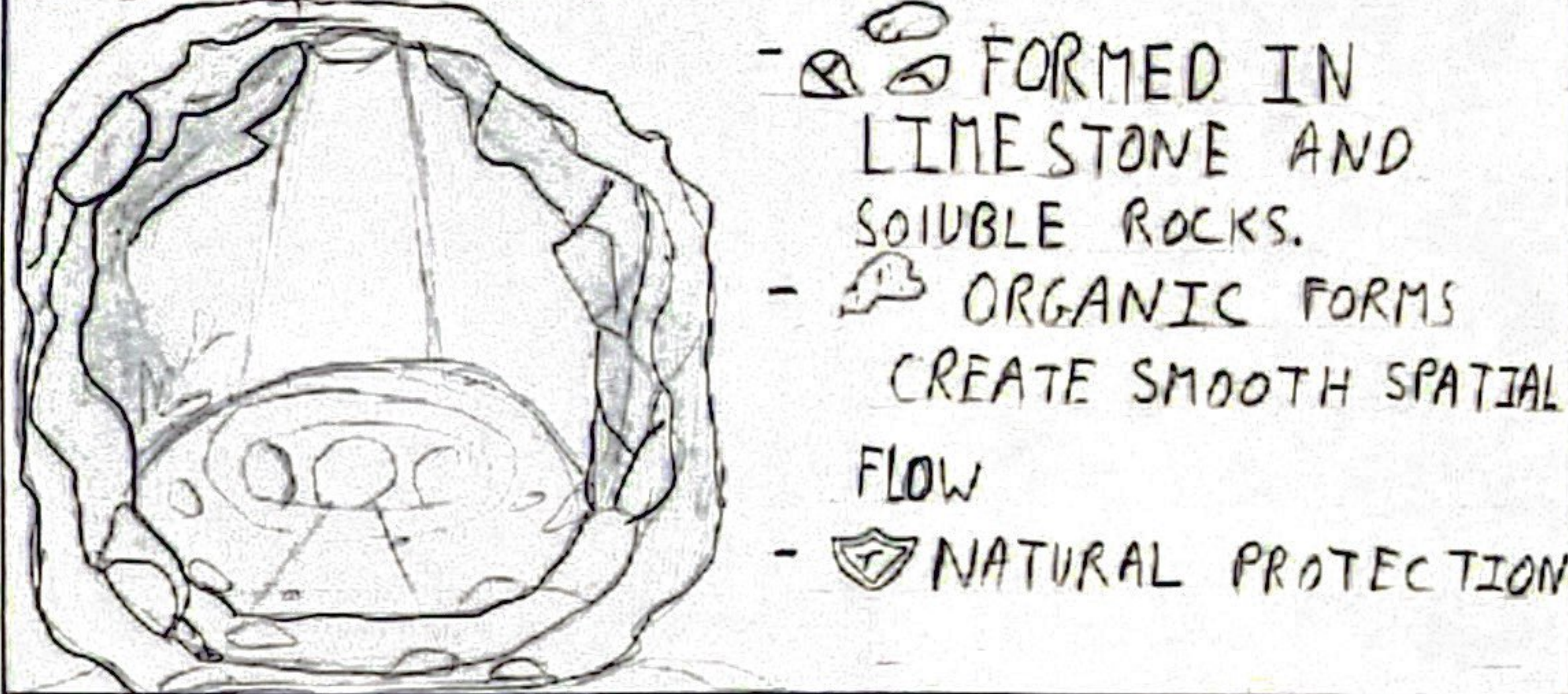
- ORGANIC
- FLUID
- FAMILY
- CONNECTION
- LIGHT
- NATURE

SITE ANALYSIS

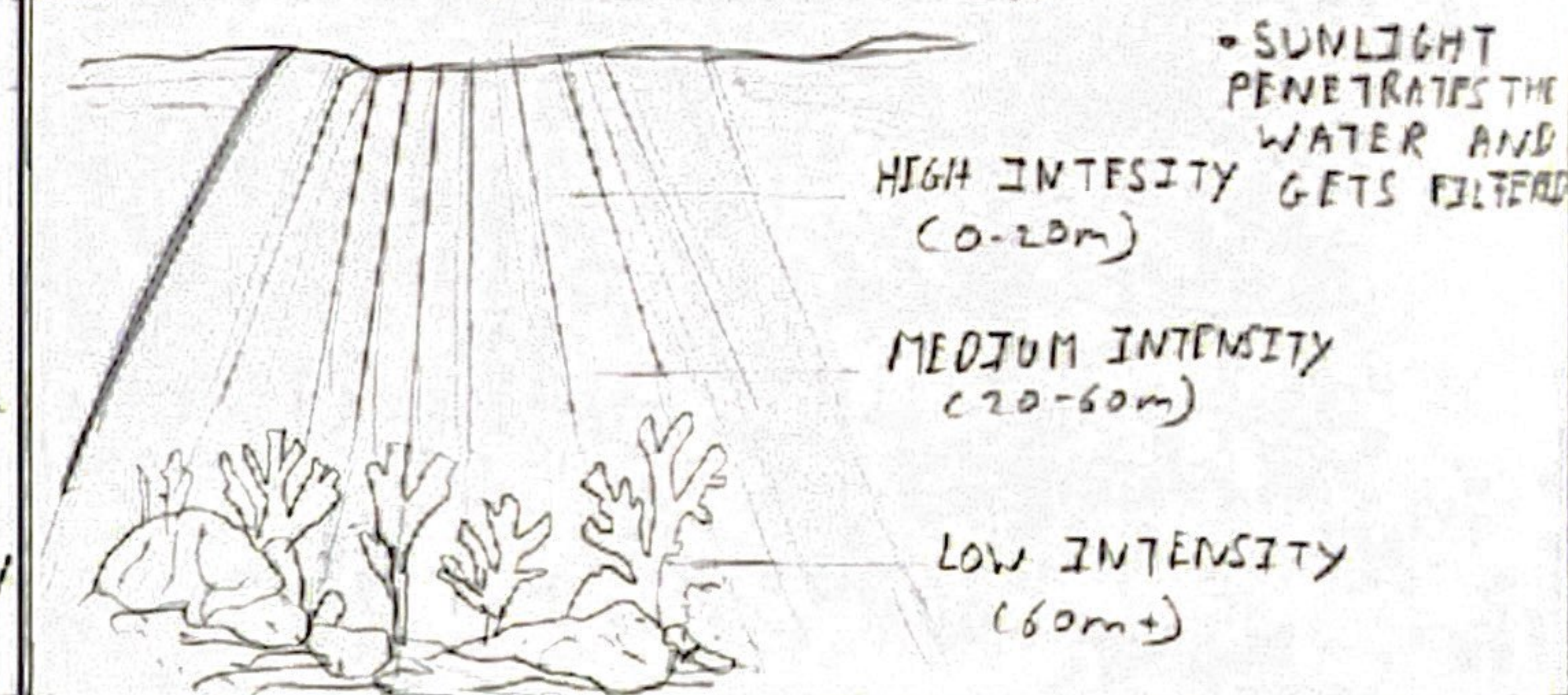
① OCEAN DEPTH



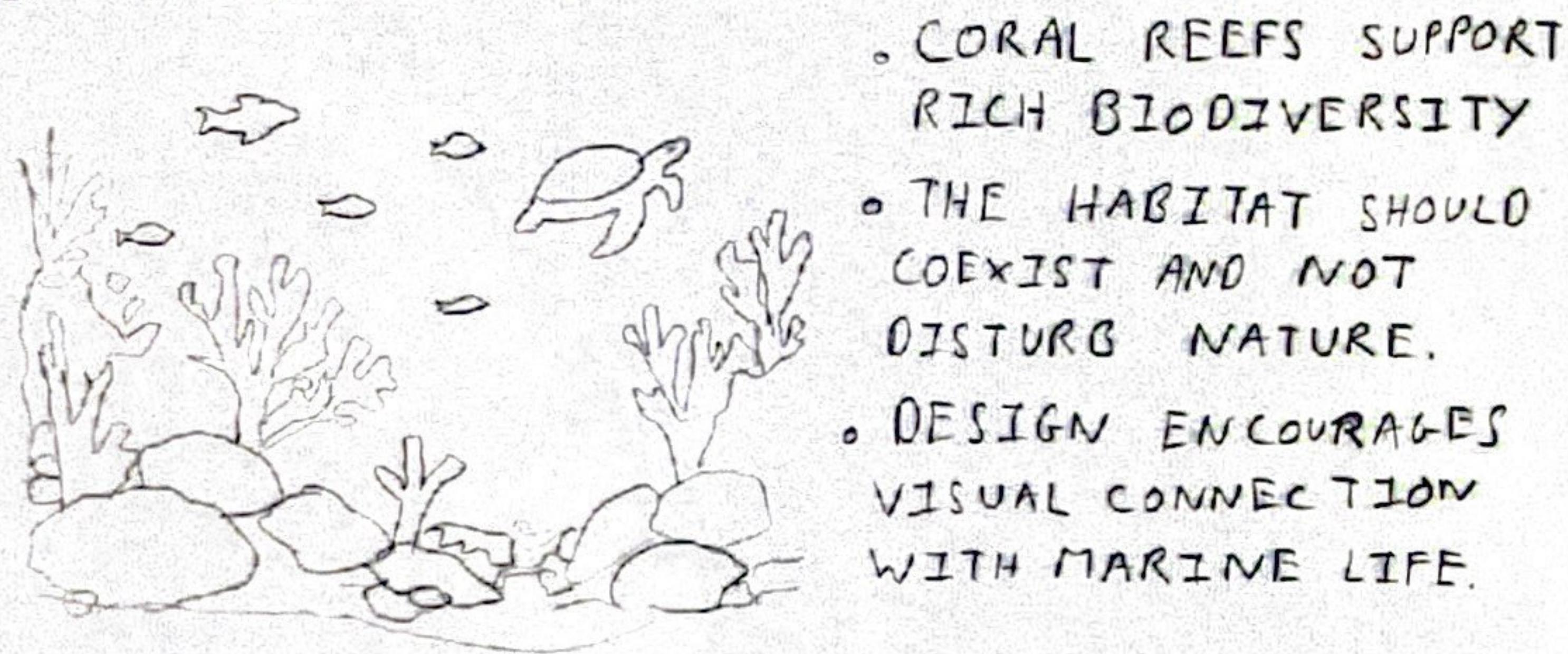
② CAVE SYSTEMS



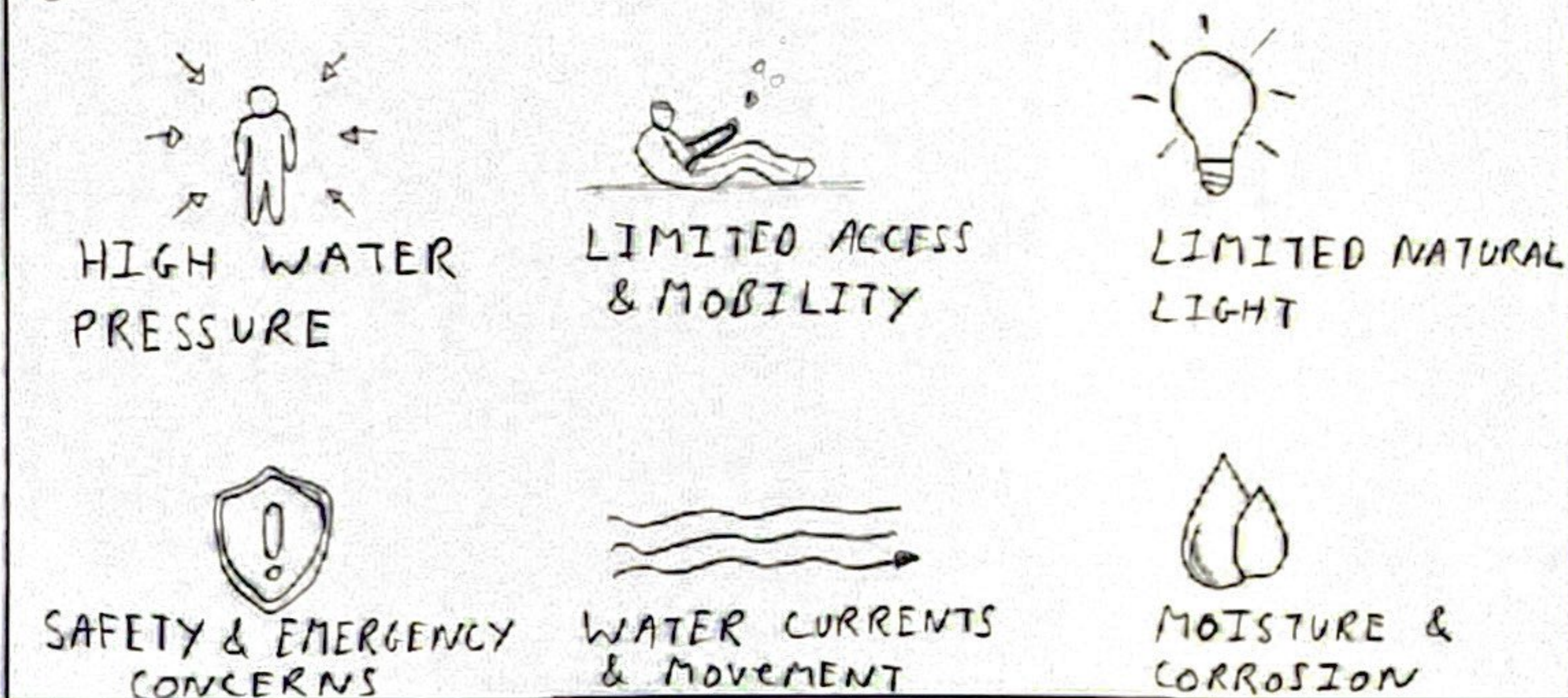
③ SUNLIGHT PENETRATION



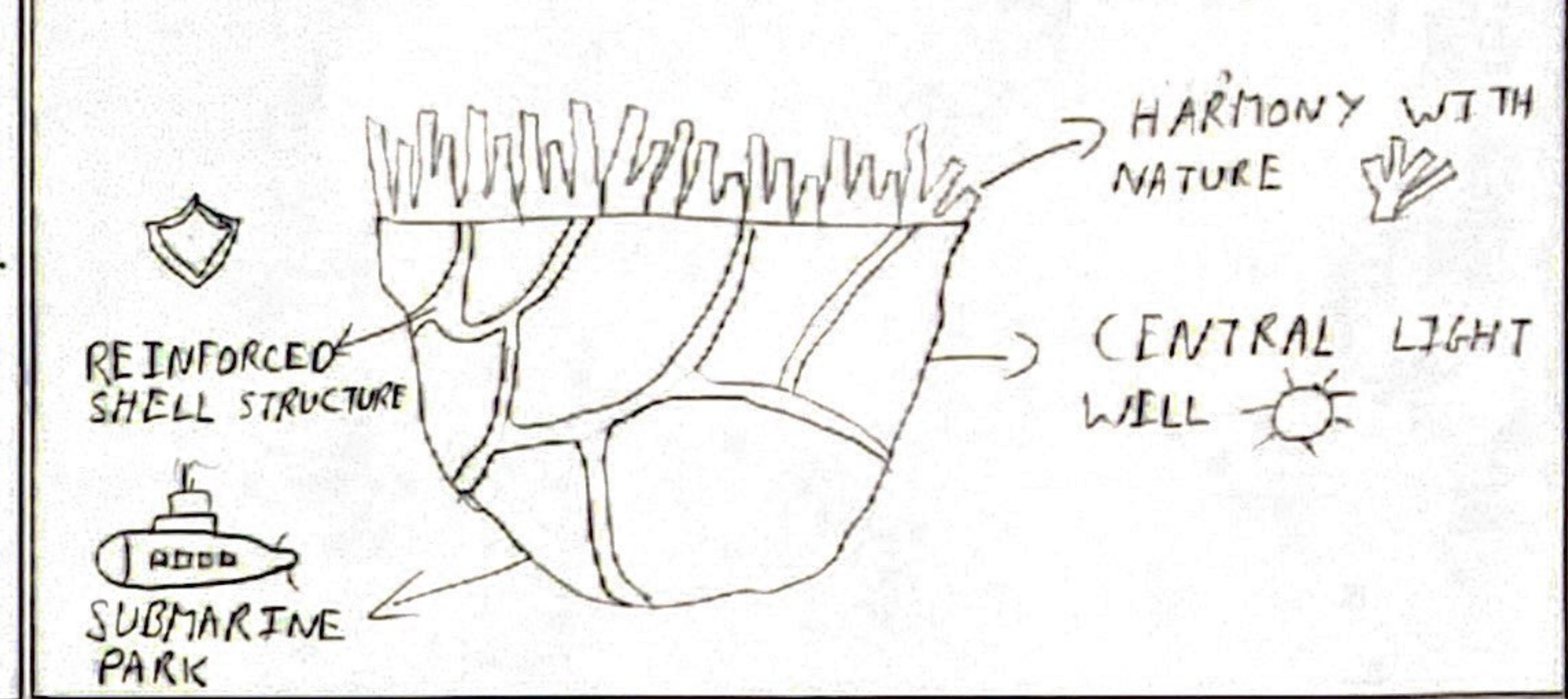
④ MARINE HABITAT



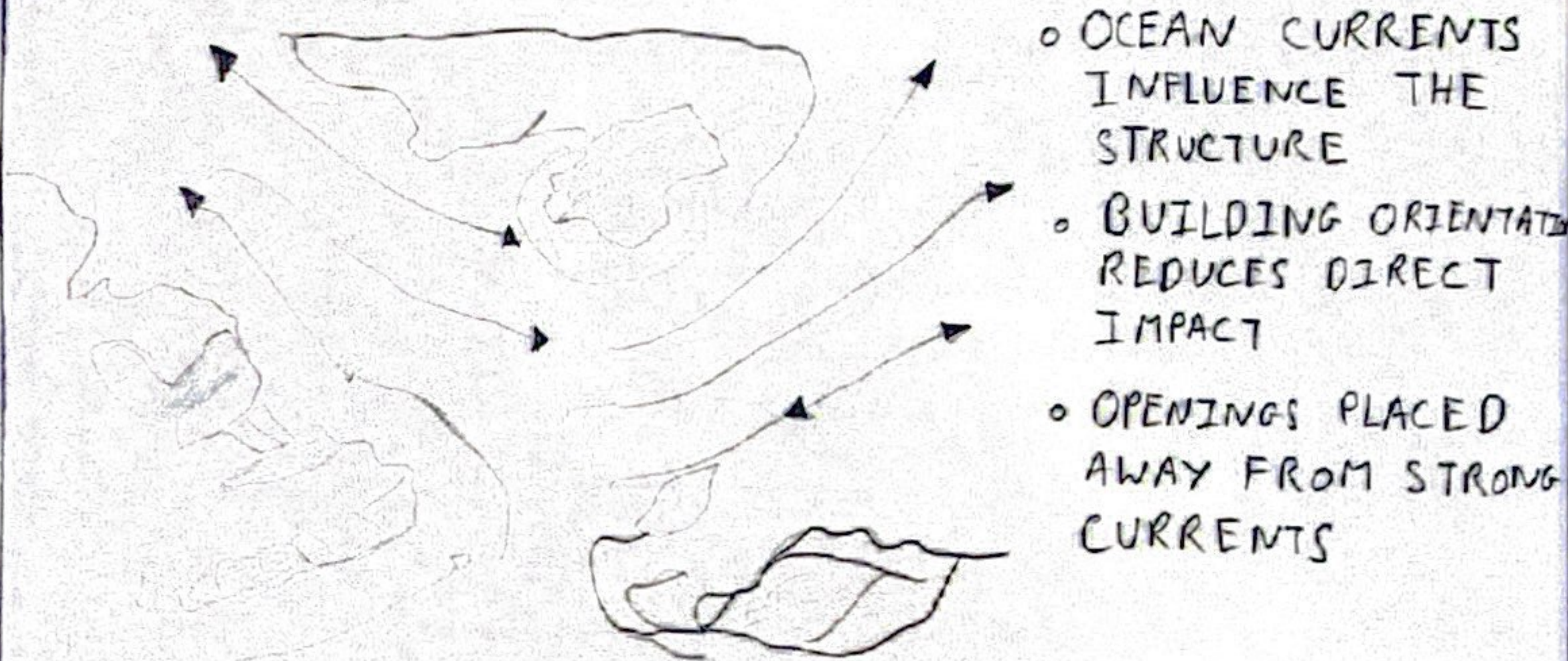
⑤ HUMAN CHALLENGES



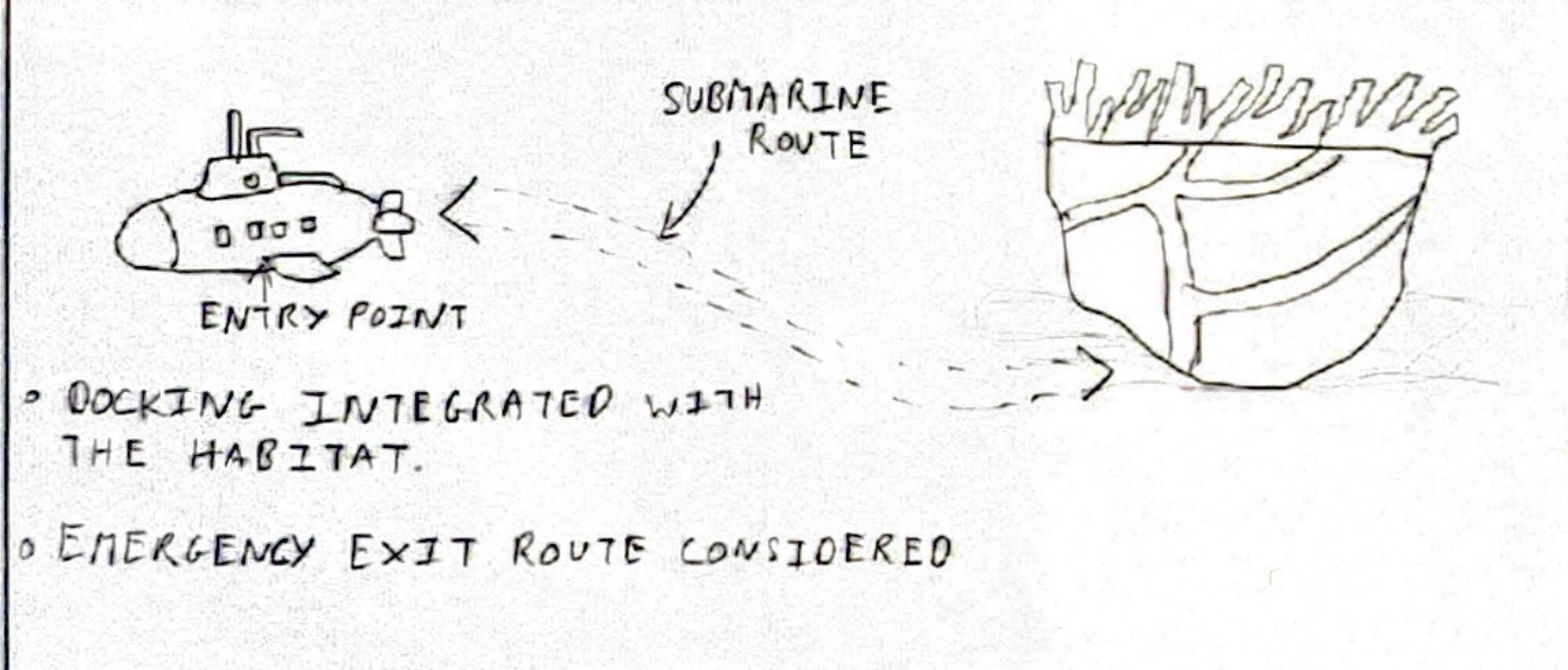
⑥ SOLUTIONS (DESIGN RESPONSES)



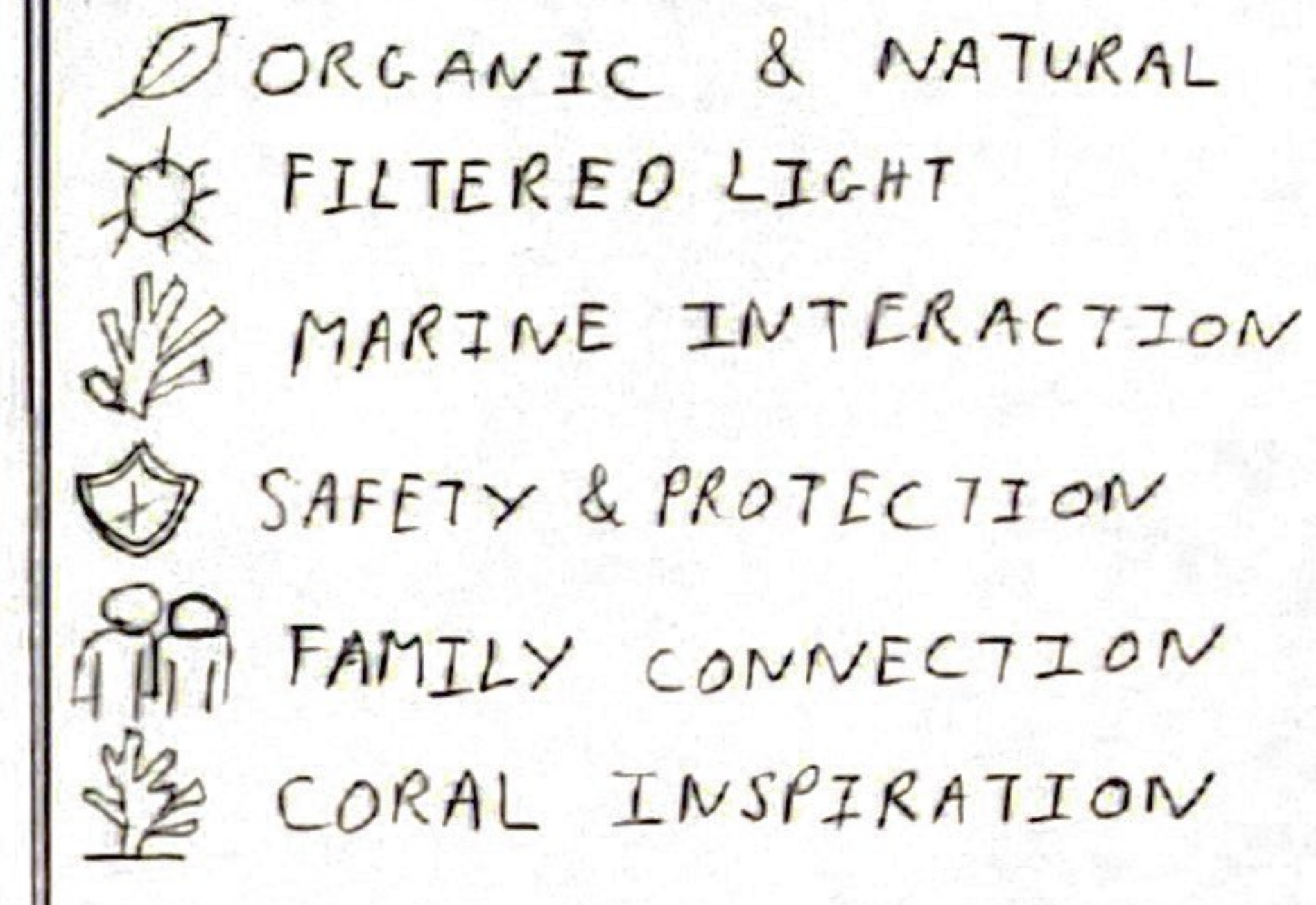
⑦ WATER CURRENT & MOVEMENT



⑧ ACCESS & CIRCULATION



⑨ KEYWORDS



SITE SUMMARY

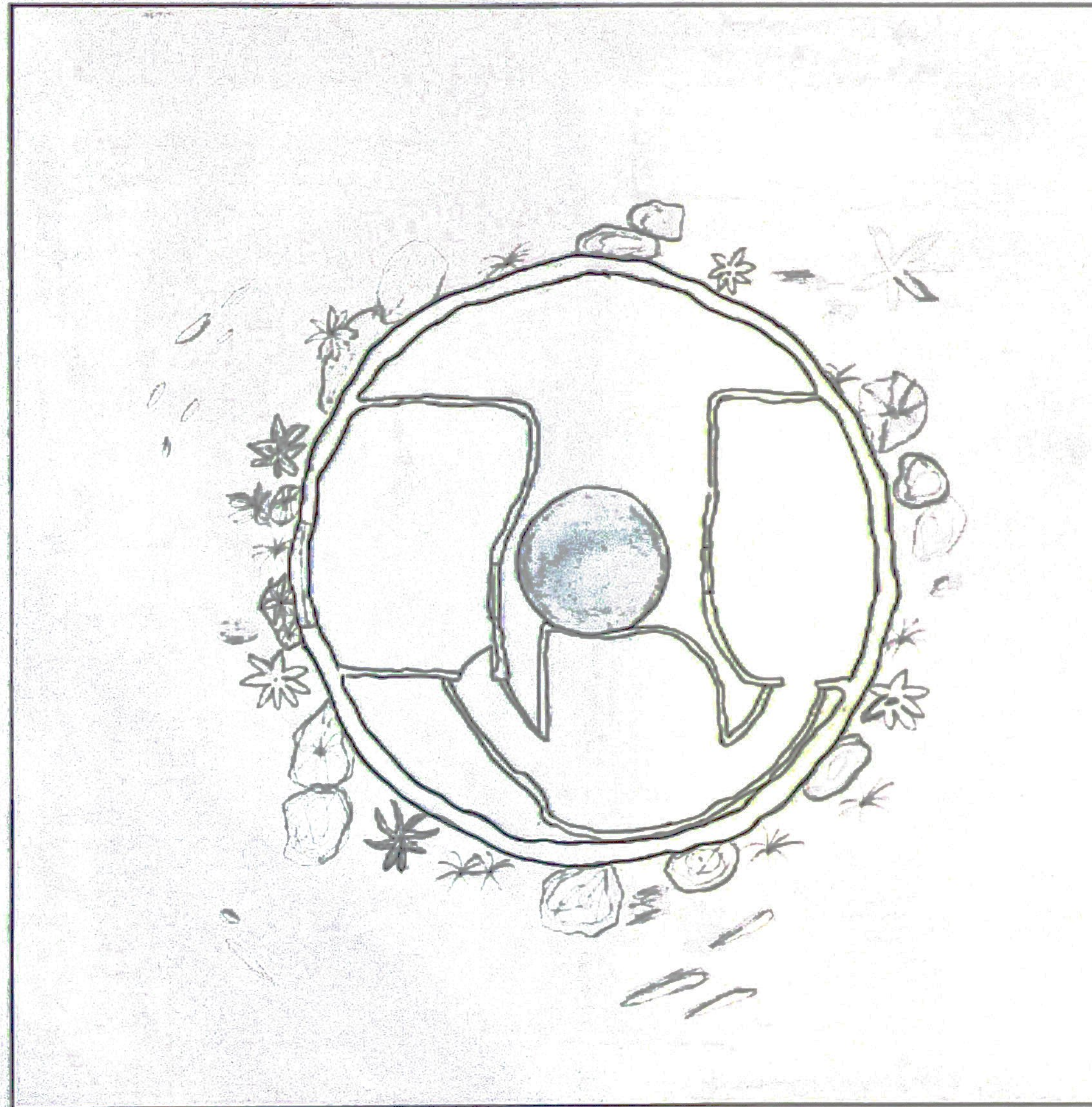
THE HOUSE IS LOCATED IN THE EPIPELAGIC (0-200m) WHERE SUNLIGHT IS AVAILABLE.

INSPIRED BY CORAL REEFS AND CAVE SYSTEM TO CREATE ORGANIC SPACES.

CHALLENGES OF THE UNDERWATER ENVIRONMENT ARE ADDRESSED THROUGH SMART DESIGN SOLUTIONS

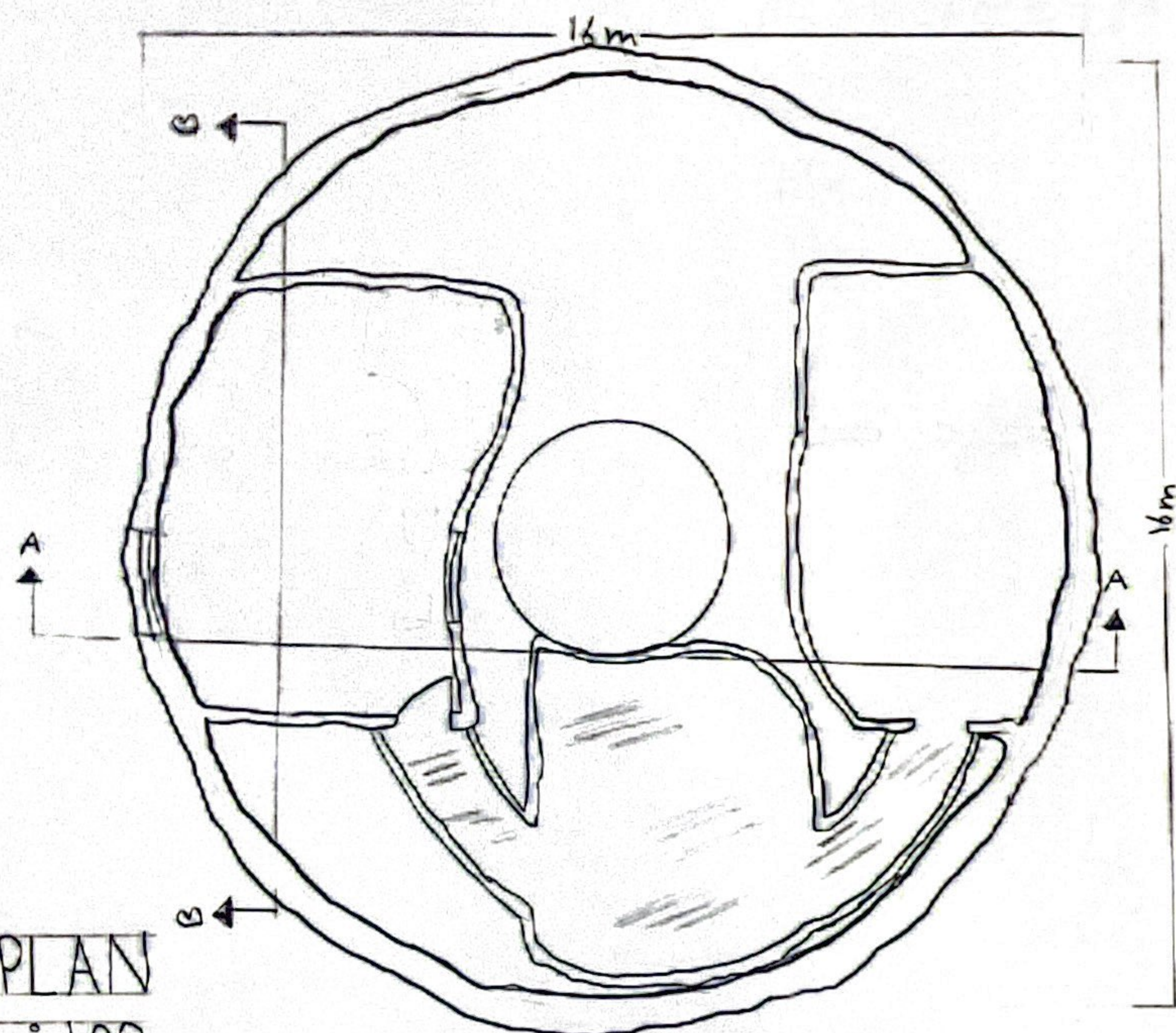
A SAFE & COMFORTABLE HOME FOR THE FAMILY UNDER OCEAN.

SITE PLAN



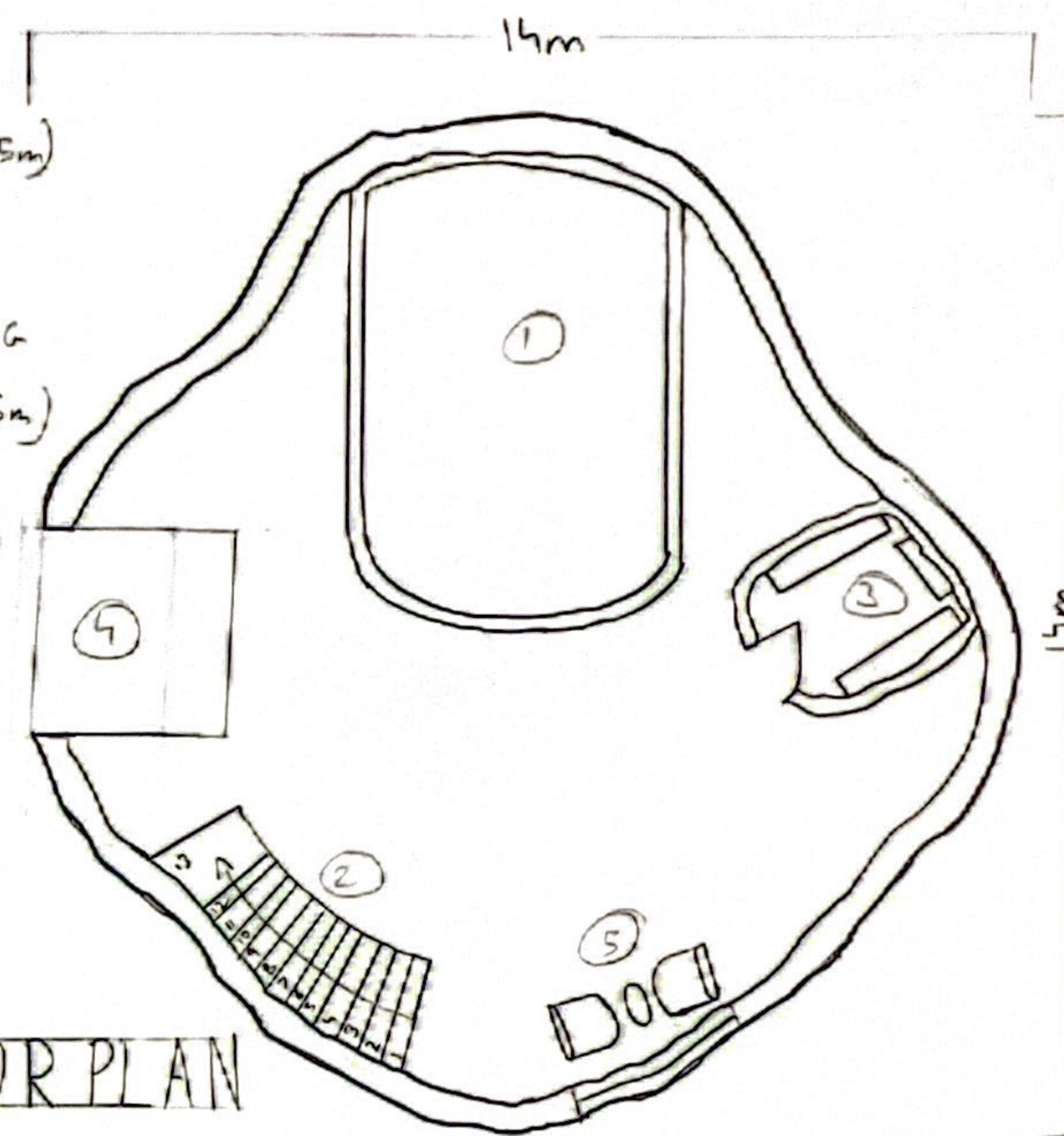
SITE INFORMATION

- SITE AREA : 10m x 50m
- TOTAL SITE AREA : 3600m²
- LOCATION : OFFSHORE
- ENVIRONMENT : UNDER WATER
- DEPTH : 200m (approx)



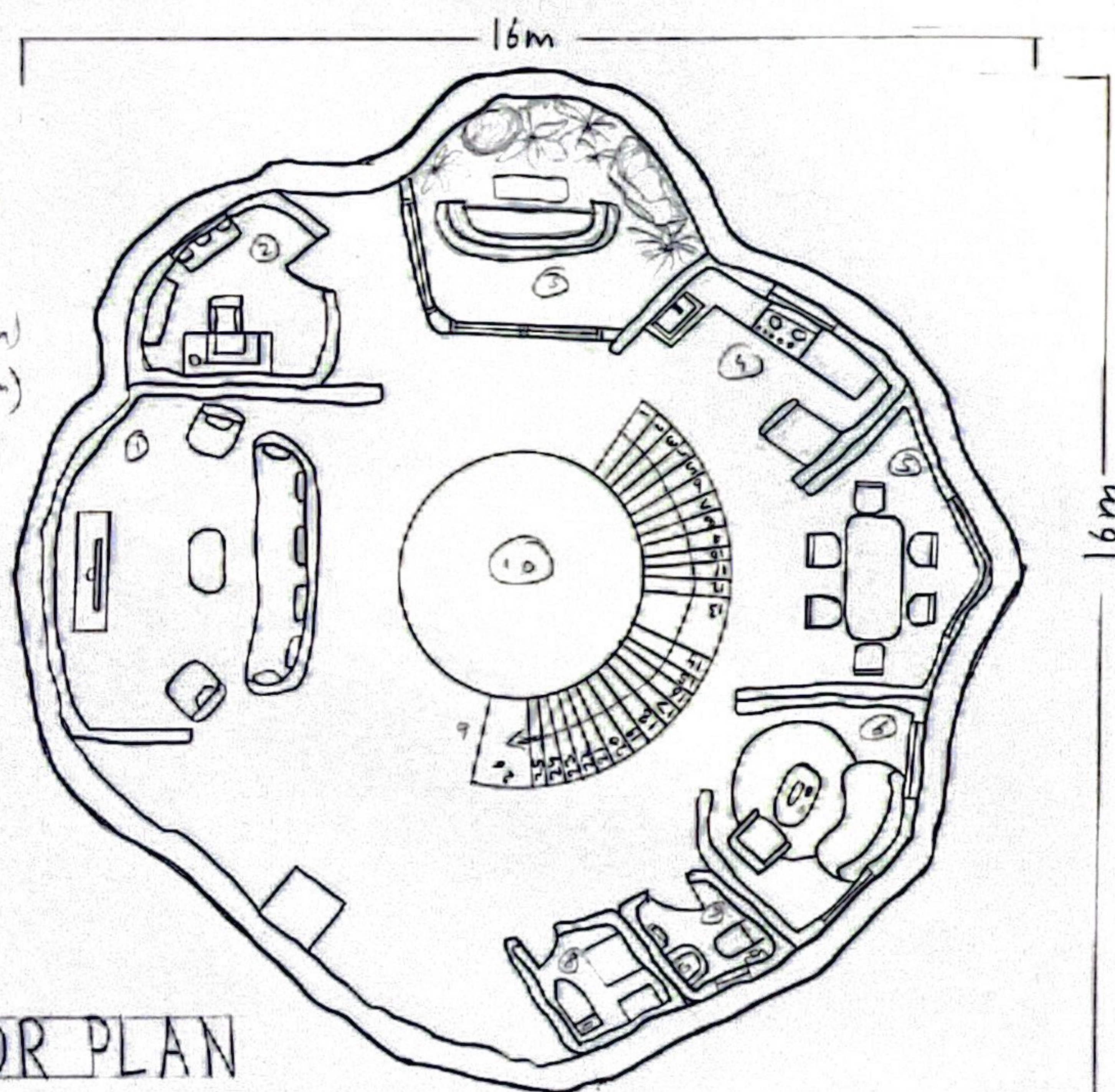
ROOF PLAN
SCALE 1:100

- ① SUBMARINE (7m x 5m)
PARKING BAY
- ② STAIRS (1500mm)
- 12 Steps & LANDING
- ③ STORAGE (2m x 2.5m)
- ④ ENTRANCE (3.5m)
- ⑤ SEATING AREA
(0.8 x 0.8m)



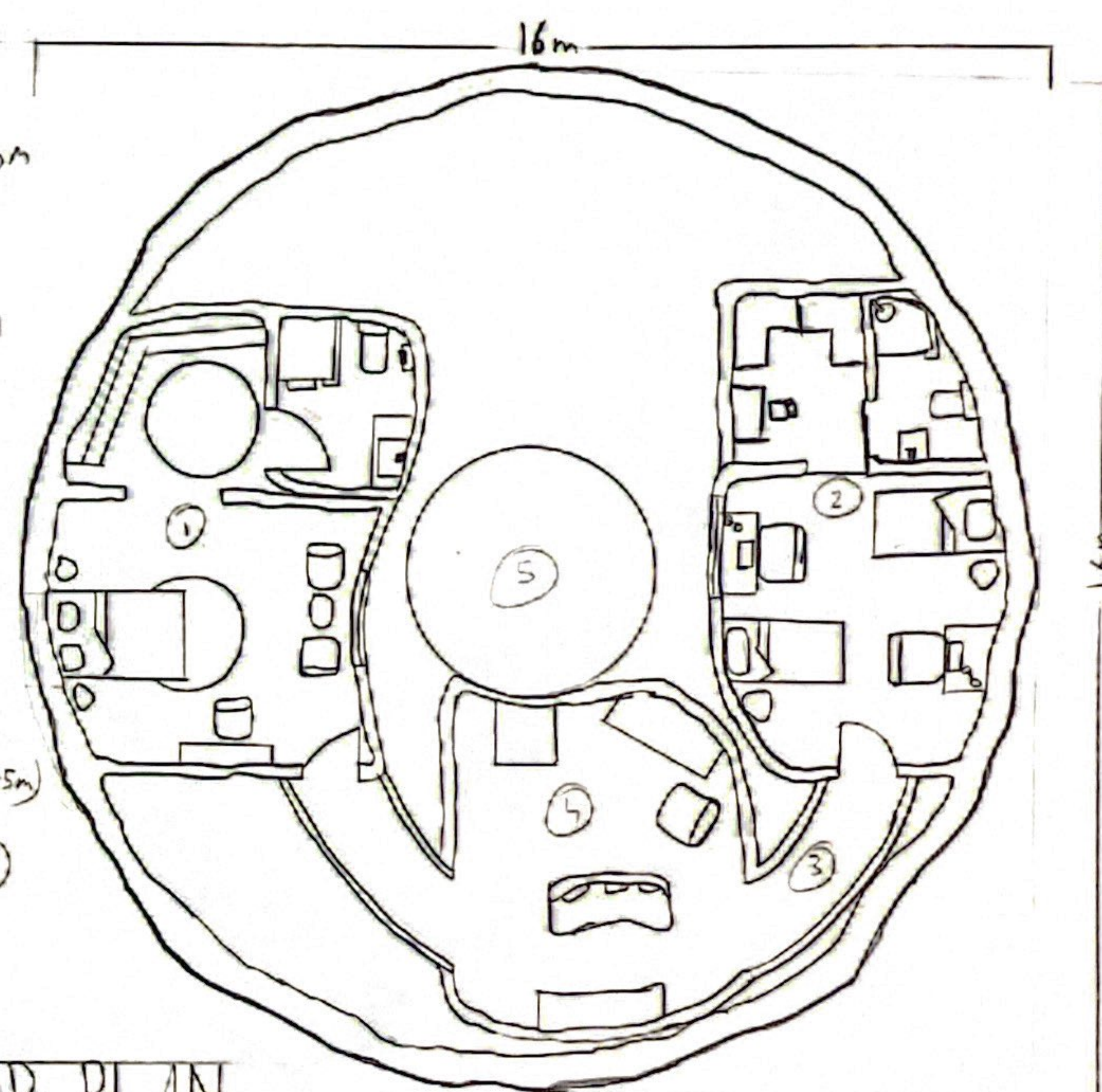
GROUND FLOOR PLAN
SCALE 1:100

- ① LIVING ROOM (6m x 4m)
- ② STUDY ROOM (3.5m x 3m)
- ③ GARDEN AREA (5m x 4m)
- ④ KITCHEN AREA (4m x 3.5m)
- ⑤ DINING AREA (5m x 3.5m)
- ⑥ DRAWING ROOM (4.5m x 4m)
- ⑦ COMMON WC (2m x 2m)
- ⑧ UTILITY ROOM (2m x 2m)
- ⑨ STAIRS - Riser (150mm)
- Tread (250mm)
- ⑩ VERTICAL
AQUARIUM (4 x 4m)

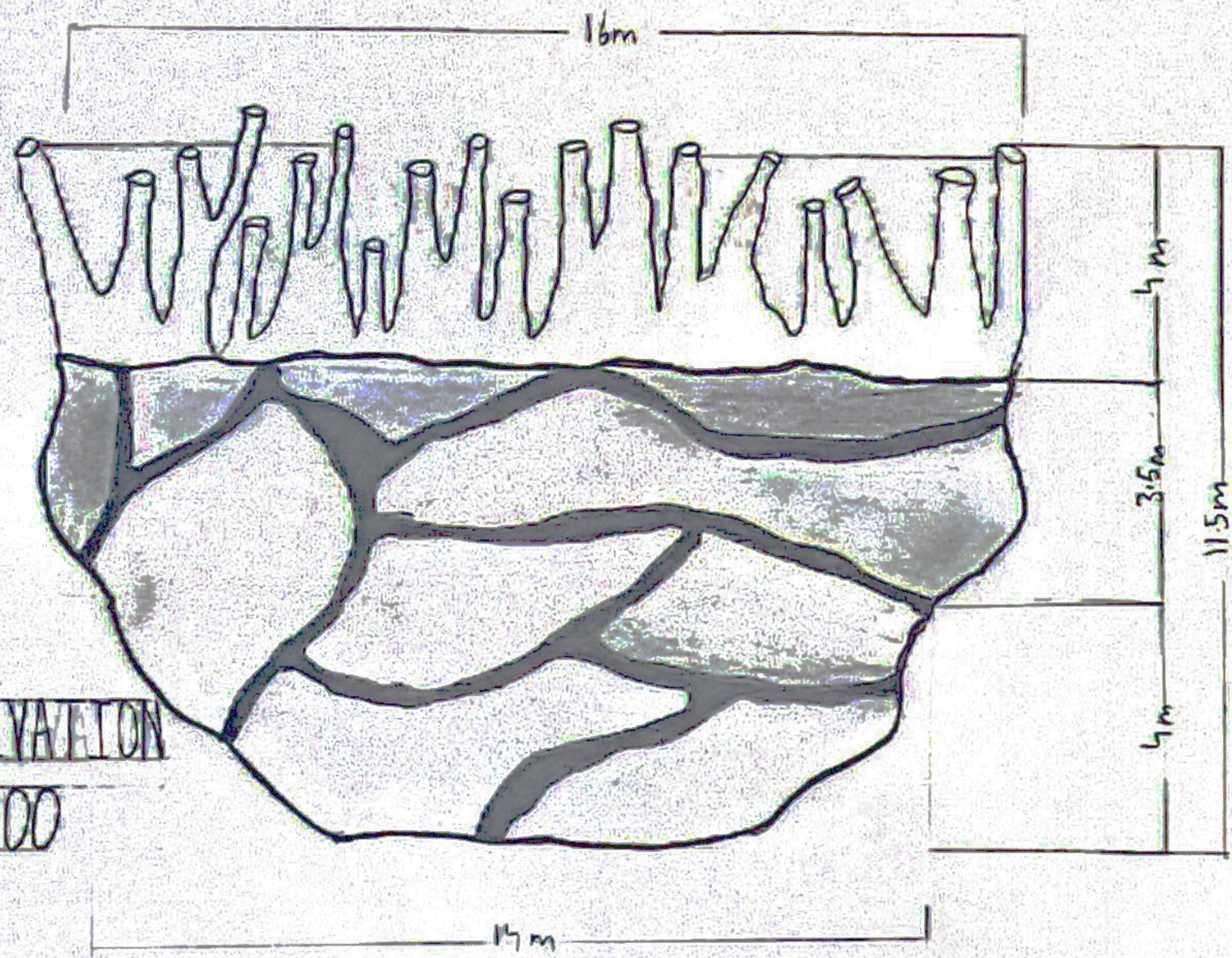


FIRST FLOOR PLAN
SCALE 1:100

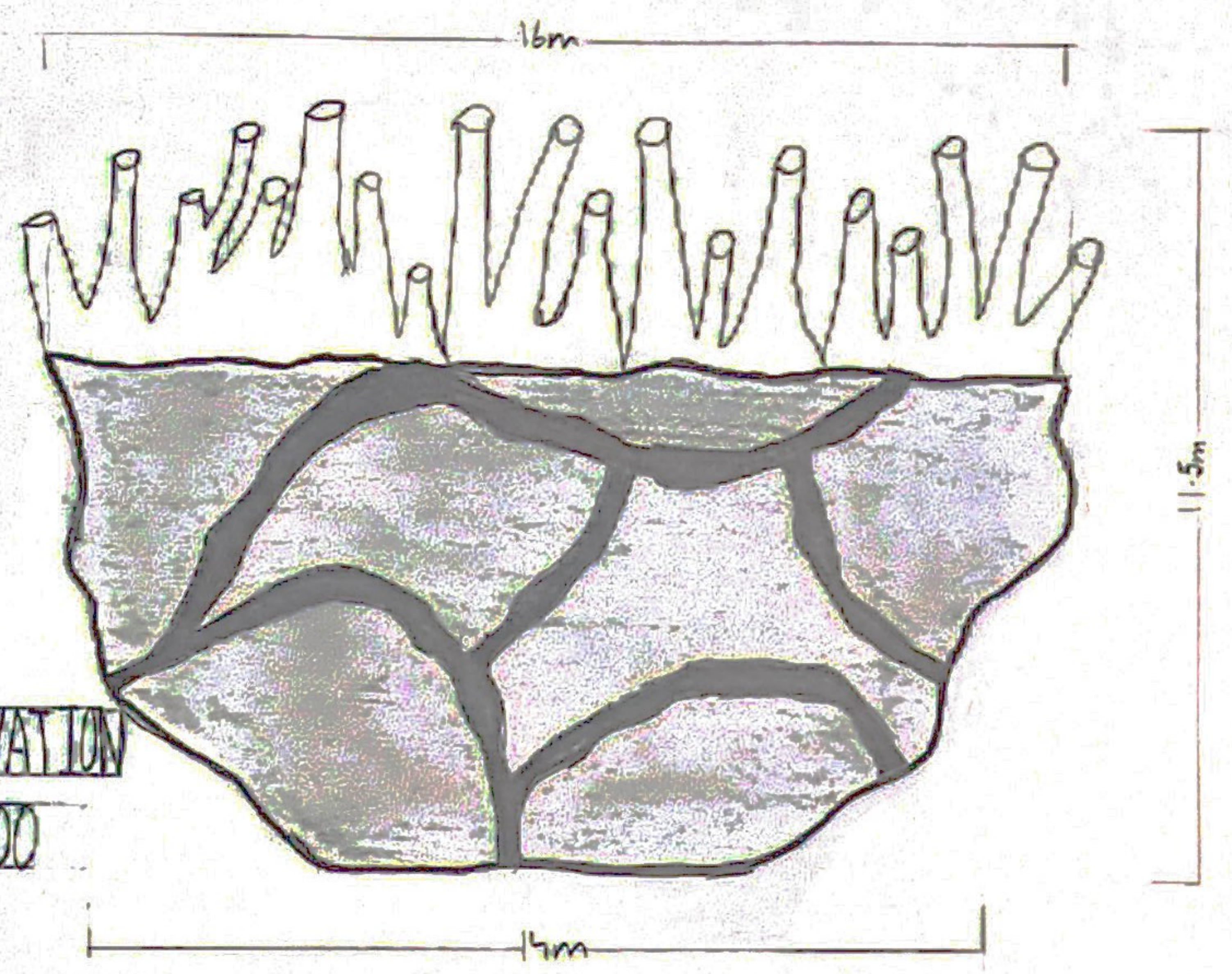
- ① BEDROOM 1 +
ATTACHED BATHROOM
- BEDROOM (5 x 5m)
- BATHROOM (2 x 3m)
- WARDROBE (2.5 x 3m)
- ② BEDROOM 2 +
ATTACHED BATHROOM
- BEDROOM (5 x 5m)
- BATHROOM (2 x 3m)
- WARDROBE (2.5 x 3m)
- ③ BRIDGE (15 x 3m)
- ④ LIVING ROOM (5m x 5m)
- ⑤ VERTICAL (4 x 4m)
AQUARIUM



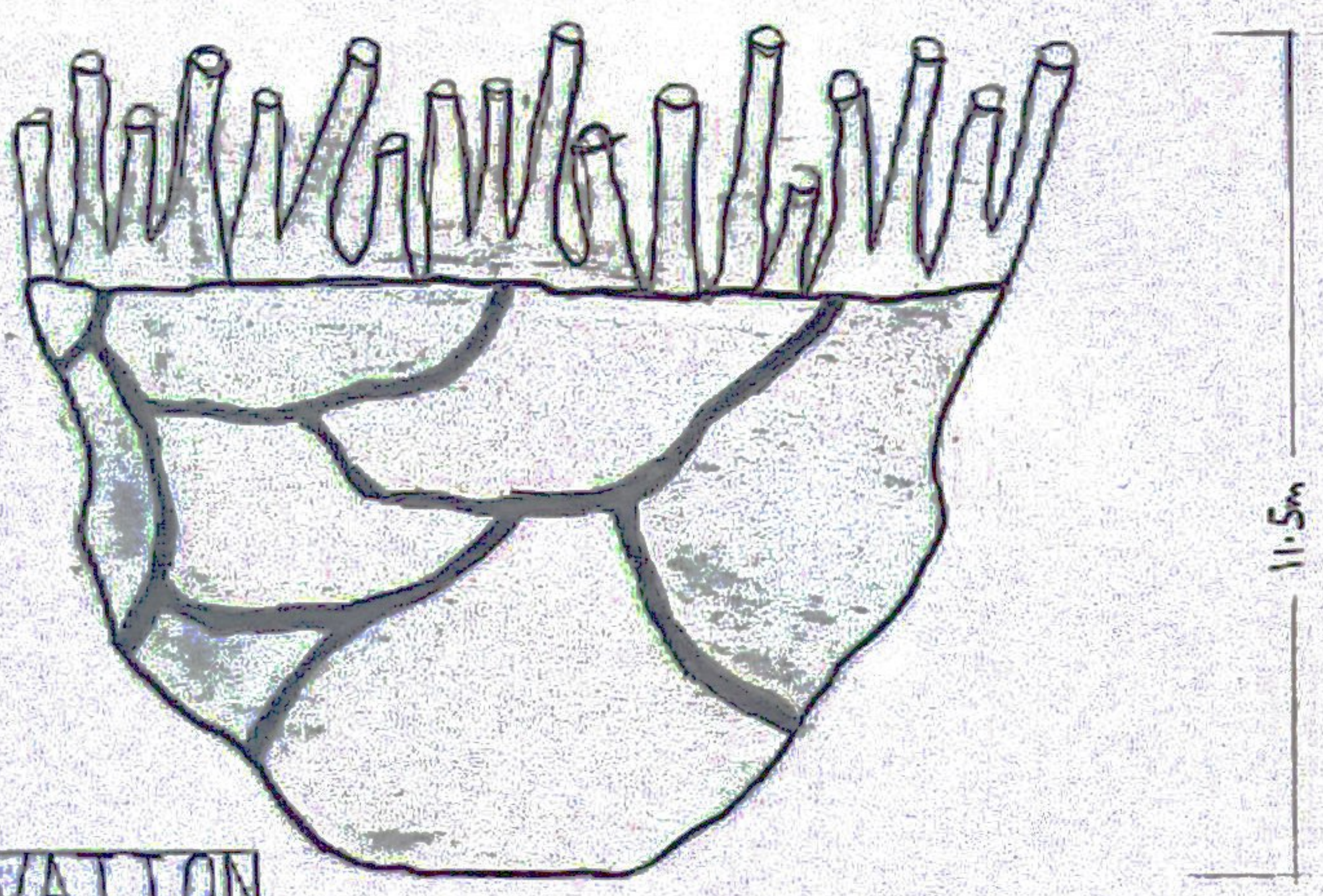
SECOND FLOOR PLAN
SCALE 1:100



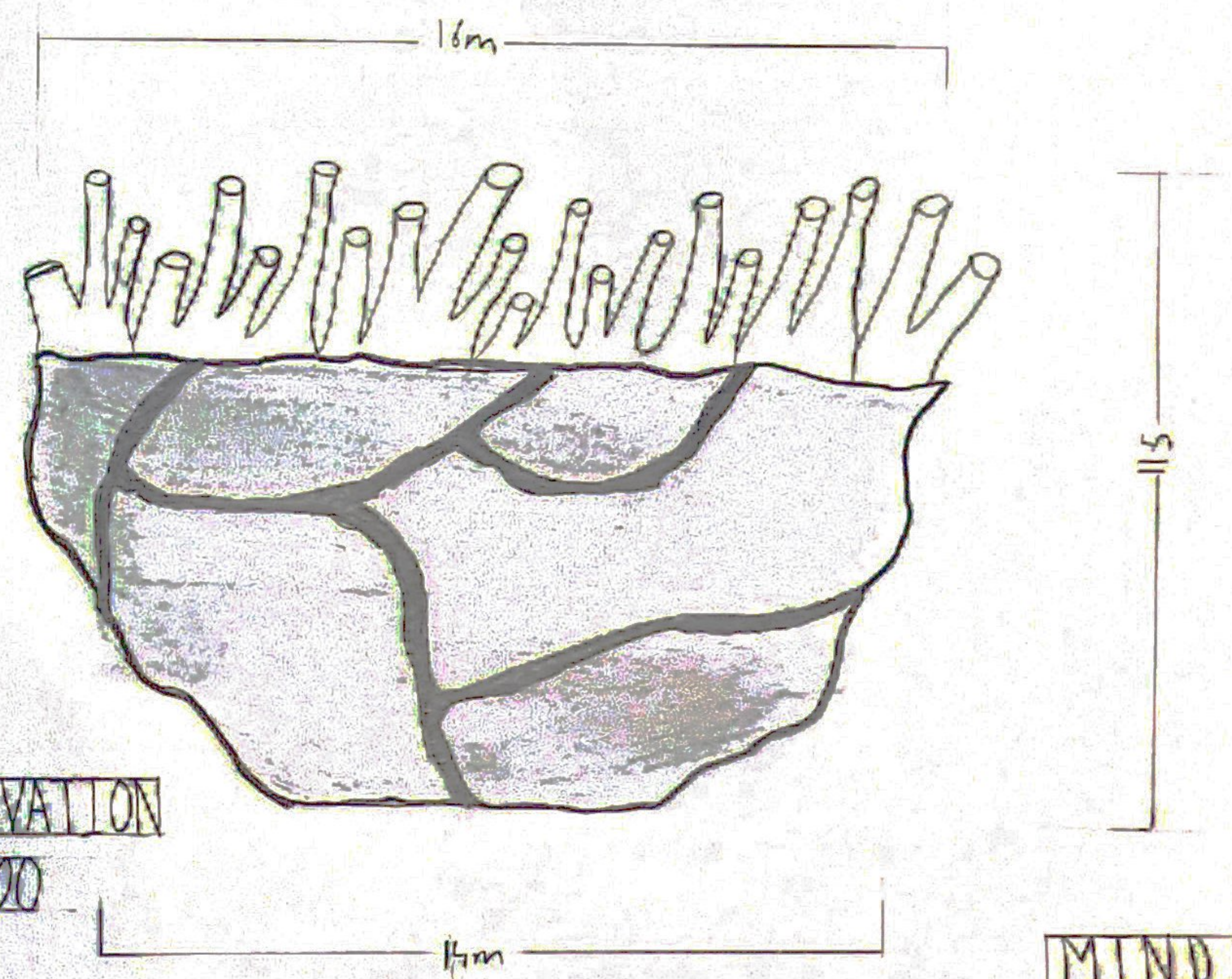
FRONT ELEVATION
SCALE 1:100



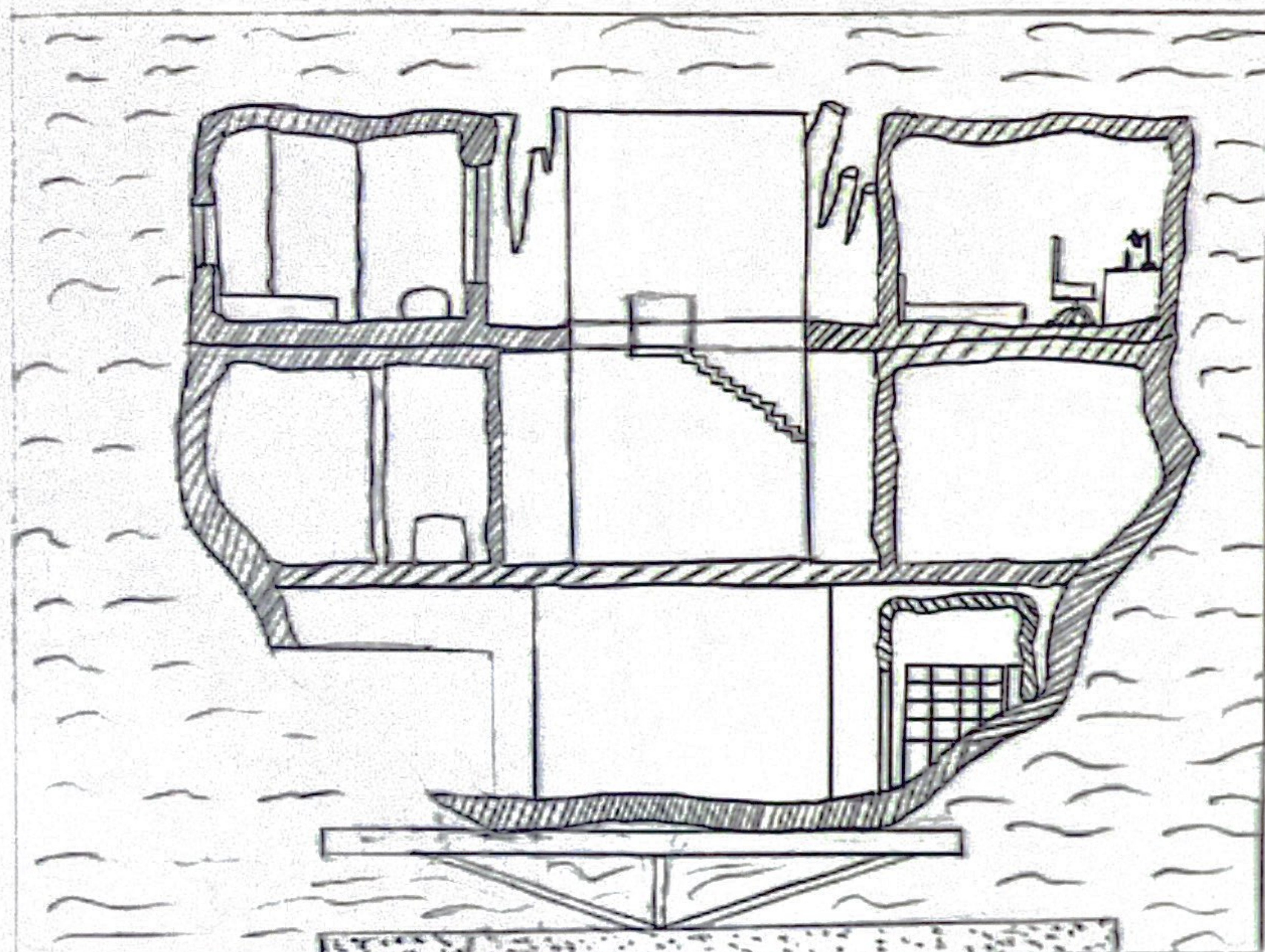
RIGHT ELEVATION
SCALE 1:100



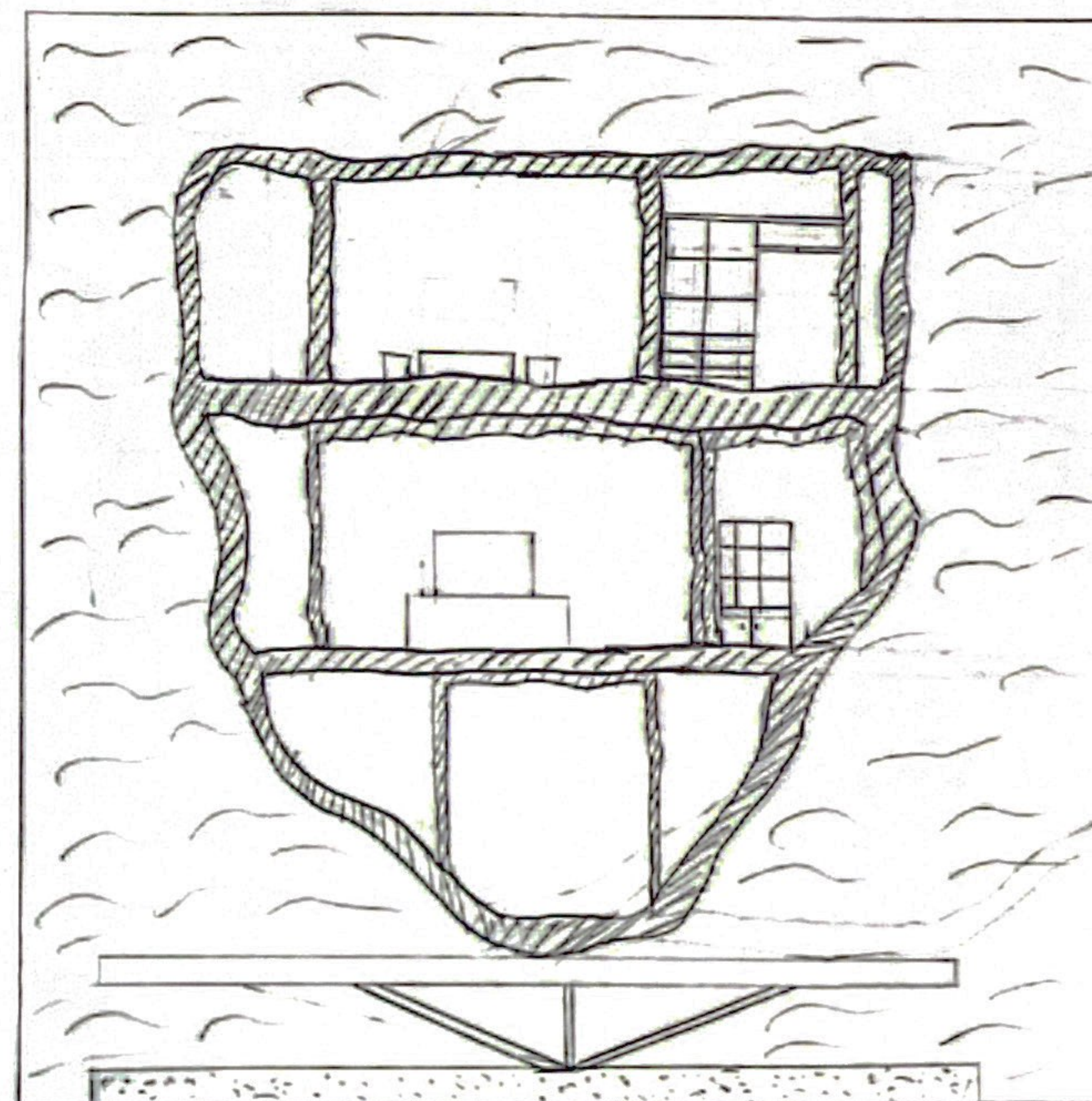
BACK ELEVATION
SCALE 1:100



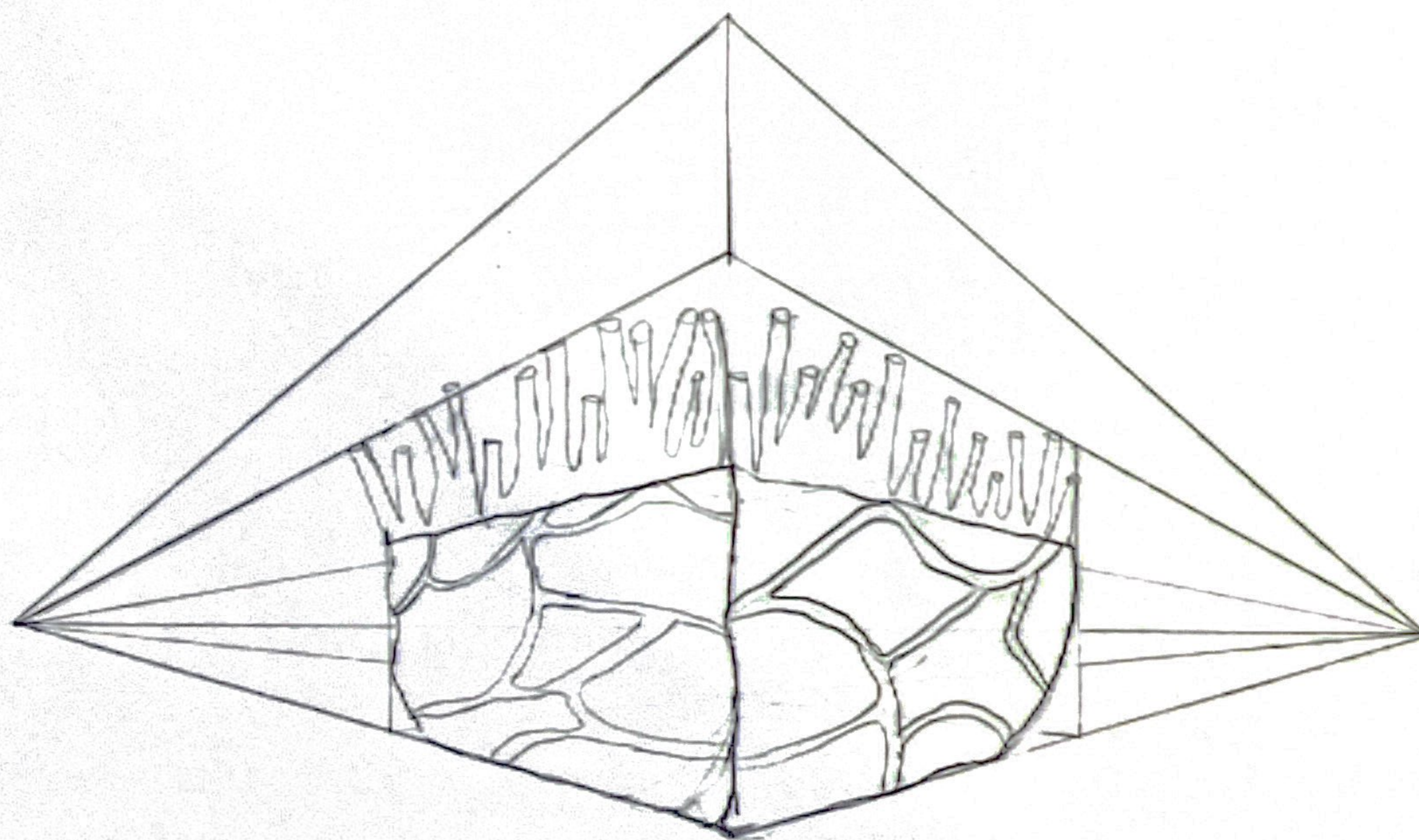
LEFT ELEVATION
SCALE 1:100



SECTION A-A
SCALE 1:100

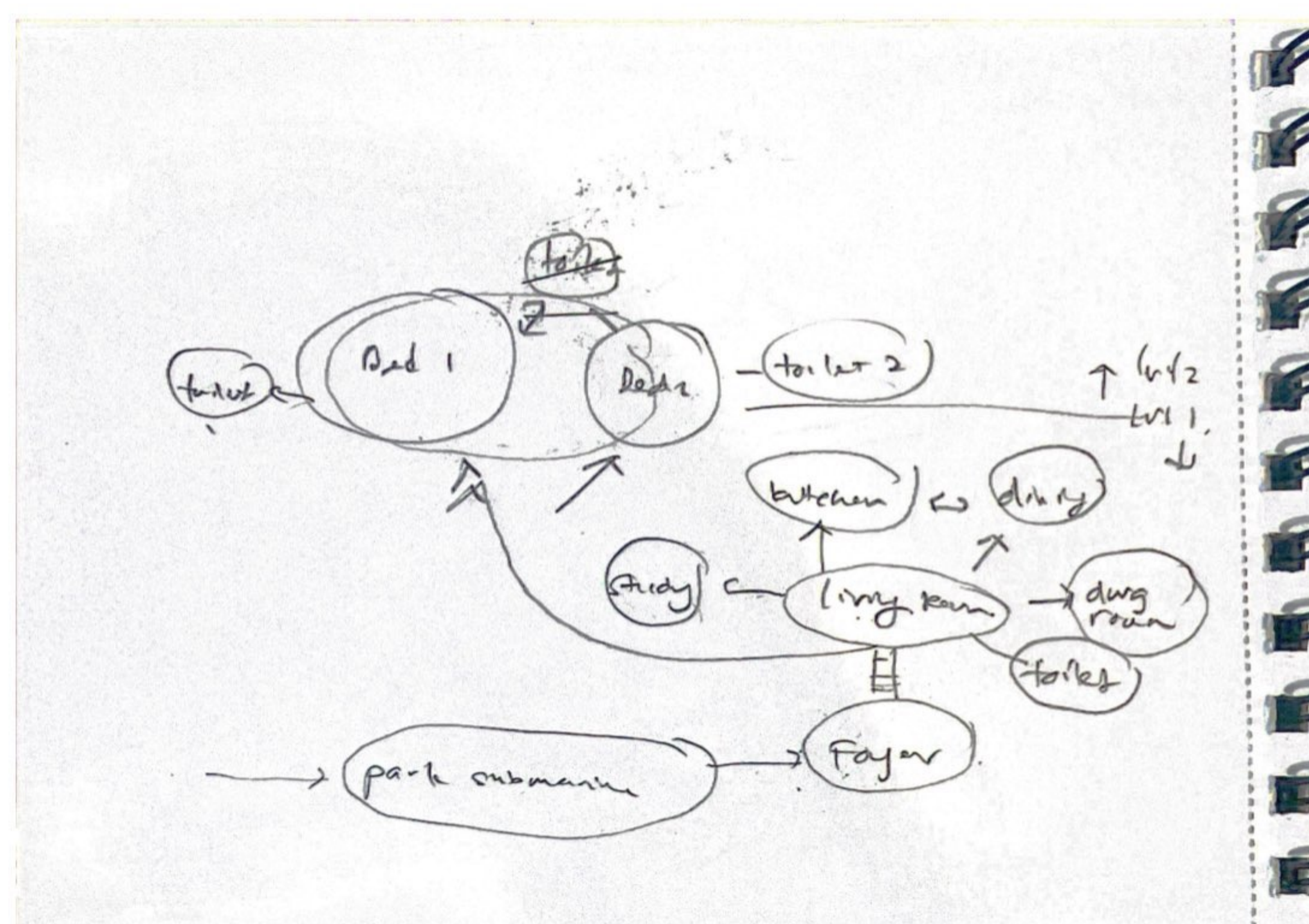
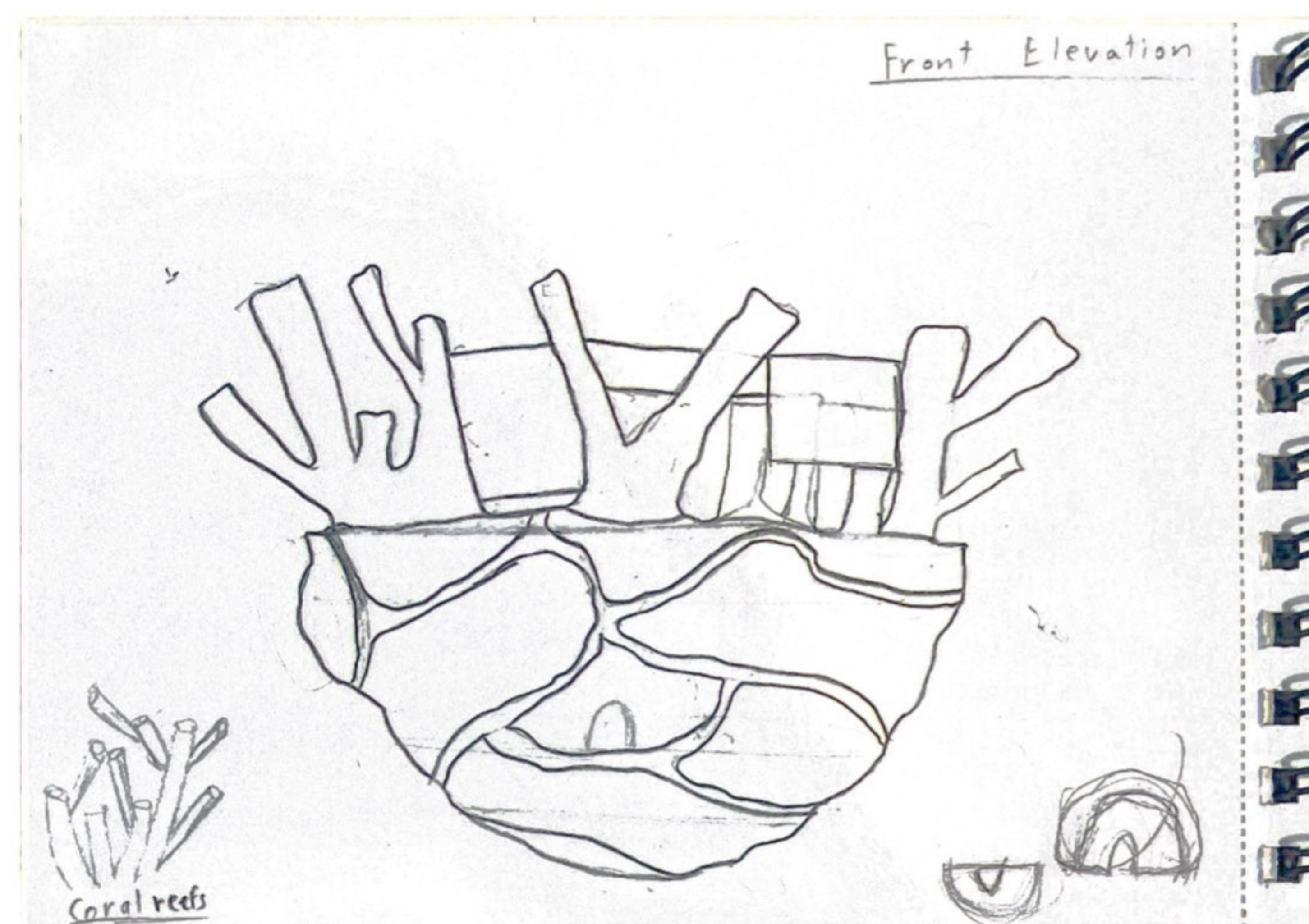
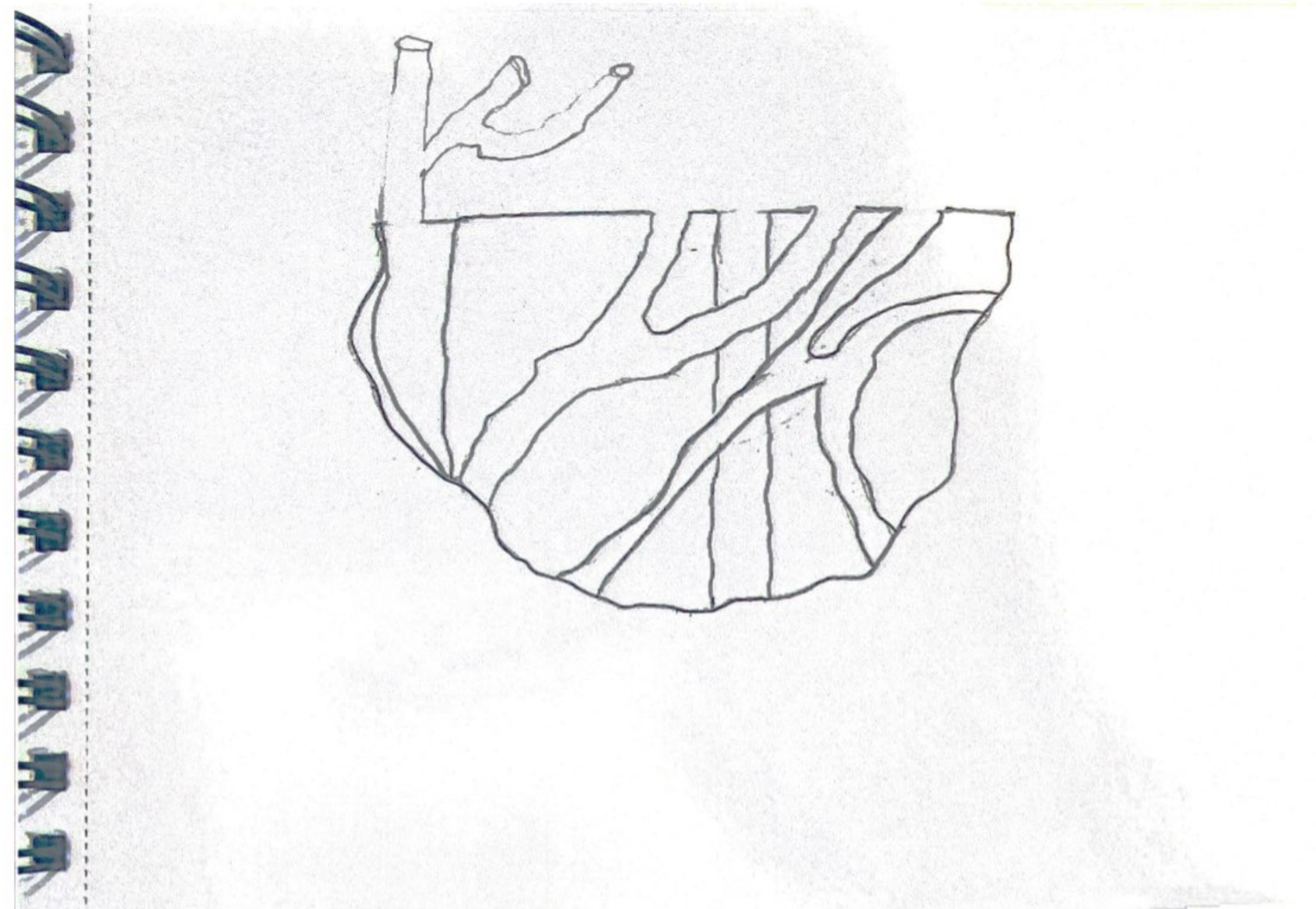
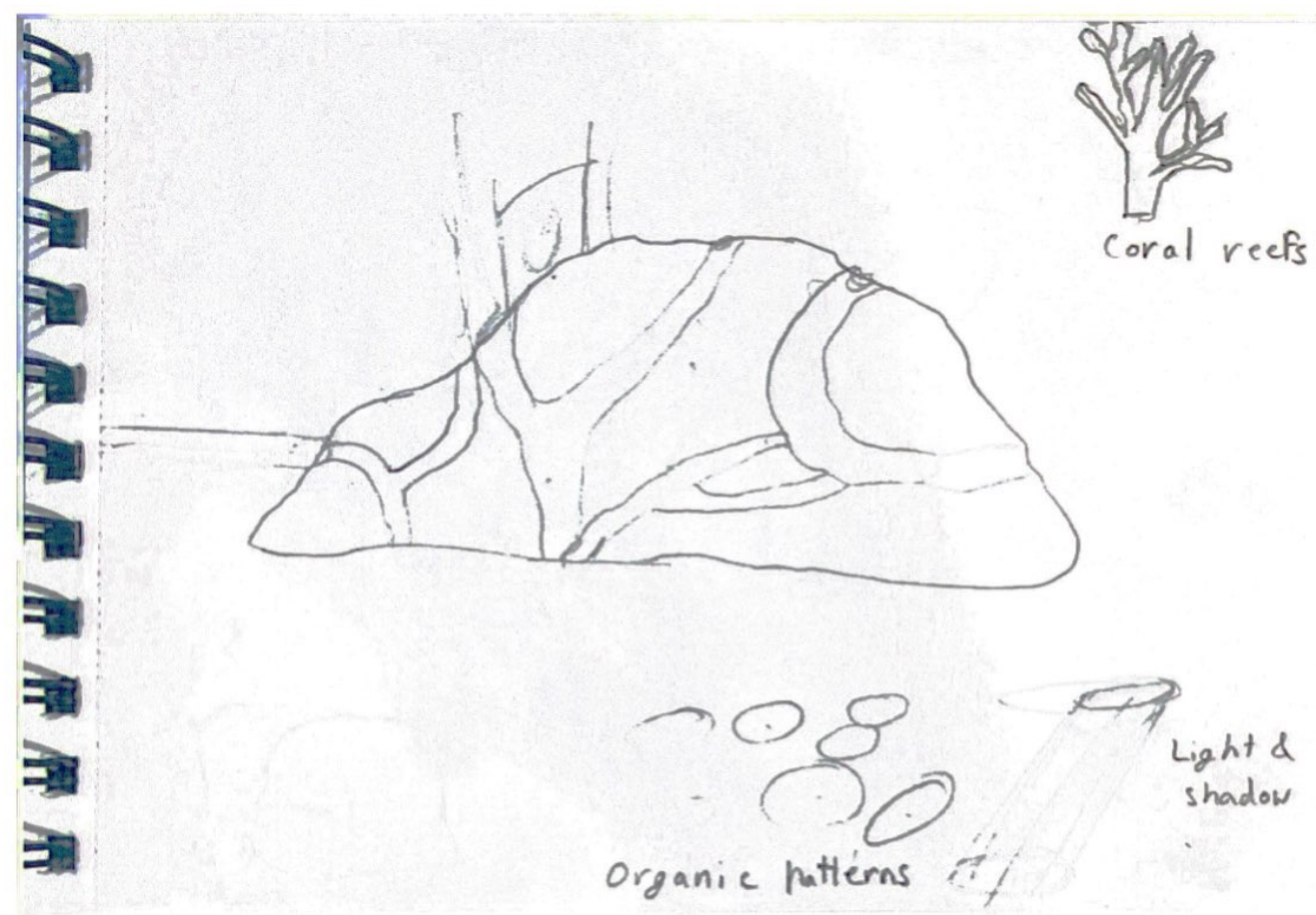


SECTION B-B
SCALE 1:100



PERSPECTIVE
SCALE 1:100

Sketches



Underwater Cave - Concept

Idea

Underwater Cave

Observation

- Sunlight
- Marine life
- Organic forms
- Coral reefs
- Shells

Concept

The design brings together filtered light & organic reefs to create a space that feels alive

Architecture

- Sunlight filtering
- Organic patterns
- Connected Space